

**Shenzhen Global Test Service Co.,Ltd.**

No.7-101 and 8A-104, Building 7 and 8, DCC Cultural and Creative Garden, No.98, Pingxin North Road, Shangmugu Community, Pinghu Street, Longgang District, Shenzhen, Guangdong

TEST REPORT
ETSI EN 300 220-1 V3.1.1 (2017-02)
ETSI EN 300 220-2 V3.2.1 (2018-06)

Report Reference No......: **GTS20211129024-1-4**

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Date of issue: Dec.30, 2021

Representative Laboratory Name: **Shenzhen Global Test Service Co.,Ltd.**

Address.....: No.7-101 and 8A-104, Building 7 and 8, DCC Cultural and Creative Garden, No.98, Pingxin North Road, Shangmugu Community, Pinghu Street, Longgang District, Shenzhen, Guangdong

Applicant's name : **ZHUHAI TAIXIN SEMICONDUCTOR CO.LTD**

Address.....: 3F, Building 2, Harbor 1 Science and Technology Park, No.1 Jin Tang Road, Tangjiawan Town, High-tech Zone, Zhuhai, Guangdong, China

Test specification..... :

Standard: ETSI EN 300 220-1 V3.1.1 (2017-02)

ETSI EN 300 220-2 V3.2.1 (2018-06)

TRF Originator: Shenzhen Global Test Service Co.,Ltd.

Master TRF: Dated 2014-12

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Test item description : **TX-AH module**

Trade Mark: N/A

Manufacturer.....: ZHUHAI TAIXIN SEMICONDUCTOR CO.LTD

Model/Type reference.....: TX-AH-R900PNR-860M

List Model: TX-AH-R900P

Ratings.....: DC 3.0-3.6V by Pinboard

Recharged by DC 5.0V

Result.....: PASS

TEST REPORT

Test Report No. :	GTS20211129024-1-4	Dec.30, 2021
		Date of issue

Equipment under Test : TX-AH module

Model /Type : TX-AH-R900PNR-860M

Listed Models : TX-AH-R900P

Applicant : **ZHUHAI TAIXIN SEMICONDUCTOR CO.LTD**

Address : 3F, Building 2, Harbor 1 Science and Technology Park, No.1 Jin Tang Road, Tangjiawan Town, High-tech Zone, Zhuhai, Guangdong, China

Manufacturer : **ZHUHAI TAIXIN SEMICONDUCTOR CO.LTD**

Address : 3F, Building 2, Harbor 1 Science and Technology Park, No.1 Jin Tang Road, Tangjiawan Town, High-tech Zone, Zhuhai, Guangdong, China

Test Result	PASS
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The test report merely corresponds to the test sample.
It is not permitted to copy extracts of these test result without the written permission of the test laboratory.

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1. TEST STANDARDS

The tests were performed according to following standards:

[ETSI EN 300 220-1 V3.1.1 \(2017-02\)](#)–Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz; Part 1: Technical characteristics and methods of measurement

[ETSI EN 300 220-2 V3.2.1 \(2018-06\)](#)–Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz; Part 2: Harmonised Standard for access to radio spectrum for non specific radio equipment

2. SUMMARY

2.1 General Remarks

Date of receipt of test sample	:	Dec.10, 2021
Testing commenced on	:	Dec.10, 2021
Testing concluded on	:	Dec.30, 2021

2.2 Product Description for Equipment Under Test (EUT)

Product Name:	TX-AH module
Trade Mark:	N/A
Model/Type reference:	TX-AH-R900PNR-860M
List Model:	TX-AH-R900P
Model Declaration	Please see description of differences
Power supply:	DC 3.0-3.6V by Pinboard Recharged by DC 5.0V
Hardware Version	V1.1
Software Version	V1.3
SRD	
Frequency Range	863.0-865.0MHz, 865.0-868.0MHz
Channel Number	1 Channel to 2M(864.0MHz, 866.0MHz) 2 Channel to 1M(863.5MHz, 864.5MHz) 3 Channel to 1M(865.5MHz, 866.5MHz, 867.5MHz)
Channel Bandwidth	1MHz, 2MHz
Modulation Type	OFDM
Antenna Description	External Antenna,3.00dBi

2.3 Difference description

The difference between TX-AH-R900PNR-860M and TX-AH-R900P series is show in the below table:

Differences / Models	TX-AH-R900P	TX-AH-R900PNR-860M
Silk screen	Different	Different
Shielding cover	Same with shielding cover	The same
Module board	/	Add a Saw to the Switch
RF IC Chip-set	TXW8301	The same
Circuit/ schematics	/	Add a Saw to the Switch
Layout	/	Add a Saw to the Switch
BOM	/	Add a Saw to the Switch
Frequency bands	802.11ah 2M:866/864; 1M:863.5/864.5/865.5/866.5/867.5	The same
BT/ Wi-Fi antenna	External Antenna	The same
Appearance	The same	The same
Dimension	The same	The same
Power Supply	The same	The same
Accessories	/	The same

2.4 Equipment Under Test

Power supply system utilised

Power supply voltage	:	☐ 230V / 50 Hz	☐ 120V / 60Hz
		☐ 12 V DC	☐ 24 V DC
		☒ Other (specified in blank below)	

DC 5.0V

2.5 Equipment Under Test

Description of the test mode

863.0-865.0MHz:

Channel	Frequency(MHz)	Channel Bandwidth (MHz)
1	863.5	1
2	864.5	1
3	864.0	2

865.0-868.0MHz:

Channel	Frequency(MHz)	Channel Bandwidth (MHz)
1	865.5	1
2	866.5	1
3	867.5	1
4	866.0	2

2.6 EUT configuration

The following peripheral devices and interface cables were connected during the measurement:

● - supplied by the manufacturer

○ - supplied by the lab

☐ Adapter	M/N:	GQ12-050200-ZG
	Manufacturer:	Dong Guan City GangQi Electronic Co.Ltd

2.7 Modifications

No modifications were implemented to meet testing criteria.

3. TEST ENVIRONMENT

3.1 Address of the test laboratory

Shenzhen Global Test Service Co.,Ltd.

No.7-101 and 8A-104, Building 7 and 8, DCC Cultural and Creative Garden, No.98, Pingxin North Road, Shangmugu Community, Pinghu Street, Longgang District, Shenzhen, Guangdong

3.2 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

CNAS (No. CNAS L8169)

Shenzhen Global Test Service Co., Ltd. has been assessed and proved to be in compliance with CNAS-CL01 Accreditation Criteria for Testing and Calibration Laboratories (identical to ISO/IEC 17025: 2019 General Requirements) for the Competence of Testing and Calibration Laboratories.

A2LA (Certificate No. 4758.01)

Shenzhen Global Test Service Co., Ltd. has been assessed by the American Association for Laboratory Accreditation (A2LA). Certificate No. 4758.01.

Industry Canada Registration Number. is 24189.

FCC Designation Number is CN1234.

FCC Registered Test Site Number is165725.

3.3 Environmental conditions

During the measurement the environmental conditions were within the listed ranges:

Normal Temperature: 25 °C

High Temperature: 45 °C

Low Temperature: -20 °C

Normal Voltage : DC 5.0V

High Voltage:DC 5.5V

Low Voltage: DC 4.5V

Relative Humidity: 55 %

Air Pressure: 989 hPa

3.4 Statement of the measurement uncertainty

The data and results referenced in this document are true and accurate. The reader is cautioned that there may be errors within the calibration limits of the equipment and facilities. The measurement uncertainty was calculated for all measurements listed in this test report acc. to TR-100028-01" Electromagnetic compatibility and Radio spectrum Matters (ERM);Uncertainties in the measurement of mobile radio equipment characteristics;Part 1" and TR-100028-02 "Electromagnetic compatibility and Radio spectrum Matters (ERM);Uncertainties in the measurement of mobile radio equipment characteristics;Part 2 " and is documented in the Shenzhen Global Test Service Co.,Ltd quality system acc. to DIN EN ISO/IEC 17025. Furthermore, component and process variability of devices similar to that tested may result in additional deviation. The manufacturer has the sole responsibility of continued compliance of the device.

Hereafter the best measurement capability for Shenzhen GTS laboratory is reported:

Item	MU	Remark
Uncertainty for Power point Conducted Emissions Test	2.42dB	
Uncertainty for Radiation Emission test in 3m chamber (30MHz to 1GHz)	3.54dB	Polarize: V
	4.10dB	Polarize: H
Uncertainty for Radiation Emission test in 3m chamber (1GHz to 25GHz)	2.08dB	Polarize: H
	2.56dB	Polarize: V

This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

3.5 SUMMARY OF TEST RESULTS

ETSI EN 300 220-2 V3.2.1 (2018-06)		
Rule	Description of Test	Result
§4.3.1	EFFECTIVE RADIATED POWER	Compliant
§4.3.2	MAXIMUM EFFECTIVE RADIATED POWER SPECTRAL DENSITY	N/A
§4.3.3	DUTY CYCLE	Compliant
§4.3.4	OCCUPIED BANDWIDTH	Compliant
§4.3.5	TX OUT OF BAND EMISSIONS	Compliant
§4.3.6	TRANSIENT POWER	Compliant
§4.3.7	ADJACENT CHANNEL POWER	Compliant
§4.3.8	TX BEHAVIOUR UNDER LOW VOLTAGE CONDITIONS	Compliant
§4.3.9	ADAPTIVE POWER CONTROL	N/A
§4.3.10	FHSS EQUIPMENT	N/A
§4.3.11	SHORT TERM BEHAVIOUR	N/A
§4.2.2	UNWANTED EMISSIONS IN THE SPURIOUS DOMAIN	Compliant
§4.4.2	BLOCKING (REVEIVER REQUIREMENTS)	Compliant
§4.2.2	RECEIVER SPURIOUS RADIATION	Compliant
§ 4.4.1	RX sensitivity(REVEIVER REQUIREMENTS)	Compliant
§ 4.5.2	Clear Channel Assessment threshold(REVEIVER REQUIREMENTS)	Compliant
§ 4.5.3	Polite spectrum access timing parameters(REVEIVER REQUIREMENTS)	Compliant

4. EFFECTIVE RADIATED POWER

4.1. Standard Applicable

According to ETSI EN 300 220-2 V3.1.1 (2017-02) section 4.3.1, The effective radiated power shall not be greater than the value allowed in annexes B or C for the chosen operational frequency band(s):

Table B.1: EU wide harmonised national radio interfaces from 25 MHz to 1 000 MHz

Operational Frequency Band		Maximum effective radiated power, e.r.p.	Maximum occupied bandwidth	Channel access and occupation rules (e.g. Duty cycle or LBT + AFA)
K	863 MHz to 865 MHz	25 mW	The whole band except for audio & video applications limited to 300 kHz	Polite spectrum access
L	865 MHz to 868 MHz	25 mW	The whole band	Polite spectrum access

NOTE 1: When either a duty cycle, Listen Before Talk (LBT) or equivalent technique applies then it shall not be user dependent/adjustable and shall be guaranteed by appropriate technical means. For LBT devices without Adaptive Frequency Agility (AFA) or equivalent techniques, the duty cycle limit applies.

4.2. Test Procedure

According to ETSI EN 300 220-1 V3.1.1 (2017-02) section 5.2.2

Effective Radiated Power (conducted measurement):

This method applies only to EUT with a permanent external antenna connector.

Test conditions:

- 1) The measurement shall be performed on the lowest and the highest Operating Frequencies declared by the manufacturer. Additional frequencies may be tested.
- 2) The transmitter shall be switched on, if possible, without modulation and the measuring receiver shall be tuned to the frequency of the transmitter under test. D-M1 test signal (unmodulated carrier) shall not be used for equipment with non-constant envelope modulation.
- 3) The RBW of the spectrum analyser shall be wide enough to cover the complete power envelope ($\geq$ OCW) of the signal of the EUT.

Measurement procedure:

The transmitter shall be connected to a dummy load as described in clause 4.3.7 and the conducted power delivered shall be measured with a measurement receiver according to clause 4.3.10.

In the case of non-constant envelope modulation, a peak detector shall be used. The maximum gain of the antenna to be used together with the equipment shall be declared by the manufacturer and this shall be recorded in the test report.

Perp, the radiated power (e.r.p.) limit applies to the maximum measured conducted power ($P_{\text{conducted}}$) value adjusted by the antenna gain (relative to a dipole) ($\text{Perp} = P_{\text{conducted}} + \text{antenna gain}$).

Effective radiated power (radiated measurement):

This measurement method applies to EUT other than those measured using clause 5.2.2.1 in ETSI EN 300 220-1 V3.1.1 (2017-02).

Test conditions:

- 1) The measurement shall be performed on the lowest and the highest Operating Frequencies declared by the manufacturer. Additional frequencies may be tested.
- 2) These measurements shall be performed at the highest power level at which the transmitter is intended to operate.
- 3) The transmitter shall be switched on, if possible, without modulation and the measuring receiver shall be tuned to the frequency of the transmitter under test. D-M1 test signal (unmodulated carrier) shall not be used for equipment with non-constant envelope modulation.
- 4) The RBW of the spectrum analyser shall be wide enough to cover the complete power envelope ($\geq$ OCW) of the signal of the EUT.
- 5) In the case of a removable antenna, the antenna shall be fitted in a manner representative of normal use.
- 6) For measurement in extreme temperature conditions, it is preferable to use an internal or a temporary connector rather than a test fixture.

Measurement procedure:

A suitable test site shall be selected from those described in clause C.1 and the radiated power established using the procedures described in clause C.5.1 (or clause C.5.2) depending on the test site, followed by clause C.5.3. In the case of non-constant envelope modulation, a peak detector shall be used.

4.3. Test Data**Environmental Conditions**

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Mode: Transmitting

863.0-865.0MHz:

Channel Bandwidth: 1MHz

Freq.	Ant.Pol.	SGO/P	Ant.	Dipole	Cable	Corrected		Limit
(MHz)	H/V	(dBm)	Gain (dB)	Gain (dB)	(dB)	power ERP		ERP
						(dBm)	(mW)	(mW)
863.5	V	2.85	10.14	0	0.64	12.35	17.18	25
863.5	H	1.63	10.13	0	0.64	11.12	12.94	25
864.5	V	2.74	10.14	0	0.64	12.24	16.75	25
864.5	H	1.65	10.13	0	0.64	11.14	13.00	25

Channel Bandwidth: 2MHz

Freq.	Ant.Pol.	SGO/P	Ant.	Dipole	Cable	Corrected		Limit
(MHz)	H/V	(dBm)	Gain (dB)	Gain (dB)	(dB)	power ERP		ERP
						(dBm)	(mW)	(mW)
864.0	V	2.63	10.14	0	0.64	12.13	16.33	25
864.0	H	1.78	10.13	0	0.64	11.27	13.40	25

865.0-868.0MHz:

Channel Bandwidth: 1MHz

Freq.	Ant.Pol.	SGO/P	Ant.	Dipole	Cable	Corrected		Limit
(MHz)	H/V	(dBm)	Gain (dB)	Gain (dB)	(dB)	power ERP		ERP
						(dBm)	(mW)	
865.5	V	2.54	10.14	0	0.64	12.04	16.00	25
865.5	H	1.86	10.13	0	0.64	11.35	13.65	25
866.5	V	2.74	10.14	0	0.64	12.24	16.75	25
866.5	H	1.67	10.13	0	0.64	11.16	13.06	25
867.5	V	2.31	10.14	0	0.64	11.81	15.17	25
867.5	H	1.89	10.13	0	0.64	11.38	13.74	25

Channel Bandwidth: 2MHz

Freq.	Ant.Pol.	SGO/P	Ant.	Dipole	Cable	Corrected		Limit
(MHz)	H/V	(dBm)	Gain (dB)	Gain (dB)	(dB)	power ERP		ERP
						(dBm)	(mW)	
866.0	V	2.47	10.14	0	0.64	11.97	15.74	25
866.0	H	1.89	10.13	0	0.64	11.38	13.74	25

Remark:

Corrected Power (dBm) = SG O/P-Cable + Ant Gain

Test Result: Compliant.

5. MAXIMUM EFFECTIVE RADIATED POWER SPECTRAL DENSITY

Not applicable: This device is neither applies to transmitters using annex B bands I, L nor using DSSS or wideband techniques other than FHSS modulation, in annex C band X. Annex B and C are in ETSI EN 300 220-2 V3.1.1 (2017-02).

6. DUTY CYCLE

6.1. Standard Applicable

In a period of 1 hour the duty cycle shall not exceed the following values given in table below.

Operational Frequency Band		Maximum effective radiated power, e.r.p.	Maximum occupied bandwidth	Channel access and occupation rules (e.g. Duty cycle or LBT + AFA)
K	863 MHz to 865 MHz	25 mW	The whole band except for audio & video applications limited to 300 kHz	Polite spectrum access
L	865 MHz to 868 MHz	25 mW	The whole band	Polite spectrum access

6.2. Test Procedure

- (1) Put the EUT on the support in its standard position and switched on.
- (2) Find the transmit signal and marker to the center frequency.
- (3) Use the zero span of spectrum analyzer to read out the “message” on and off time in any one period.

6.3. Test Data

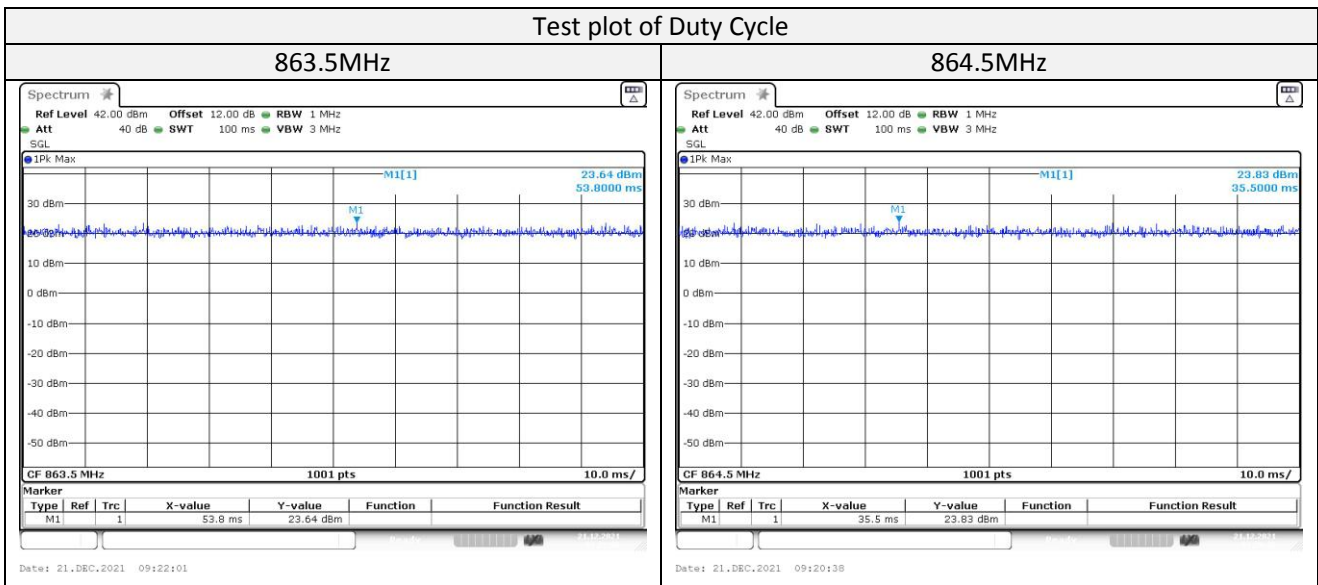
863.0-865.0MHz:

Channel Bandwidth: 1MHz

Frequency (MHz)	Transmit On time (ms)	Duty Cycle	Limit
863.5	100	100%	/
864.5	100	100%	/

Note: The duty cycle is calculated as follow:

$$\text{Duty cycle} = \text{Transmit On time (s)} / (\text{Within 1 hour}) / 3600(\text{s}) * 100\%$$

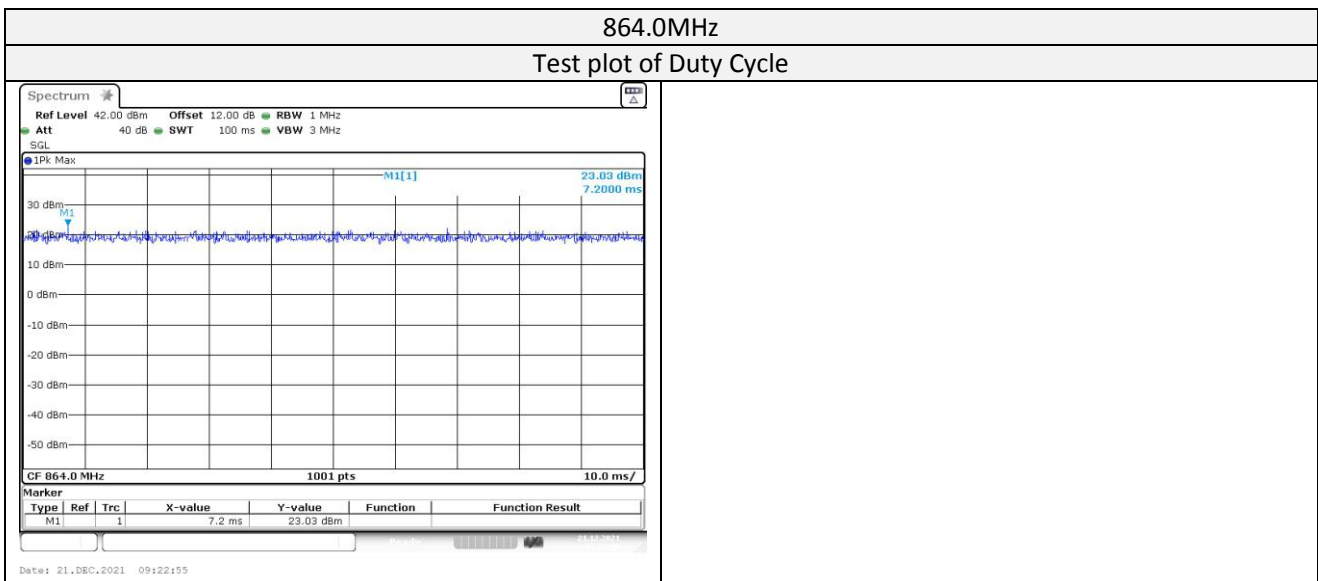


Channel Bandwidth: 2MHz

Frequency (MHz)	Transmit On time (ms)	Duty Cycle	Limit
864.0	100	100%	/

Note: The duty cycle is calculated as follow:

$$\text{Duty cycle} = \text{Transmit On time (s)} / (\text{Within 1 hour}) / 3600(\text{s}) * 100\%$$



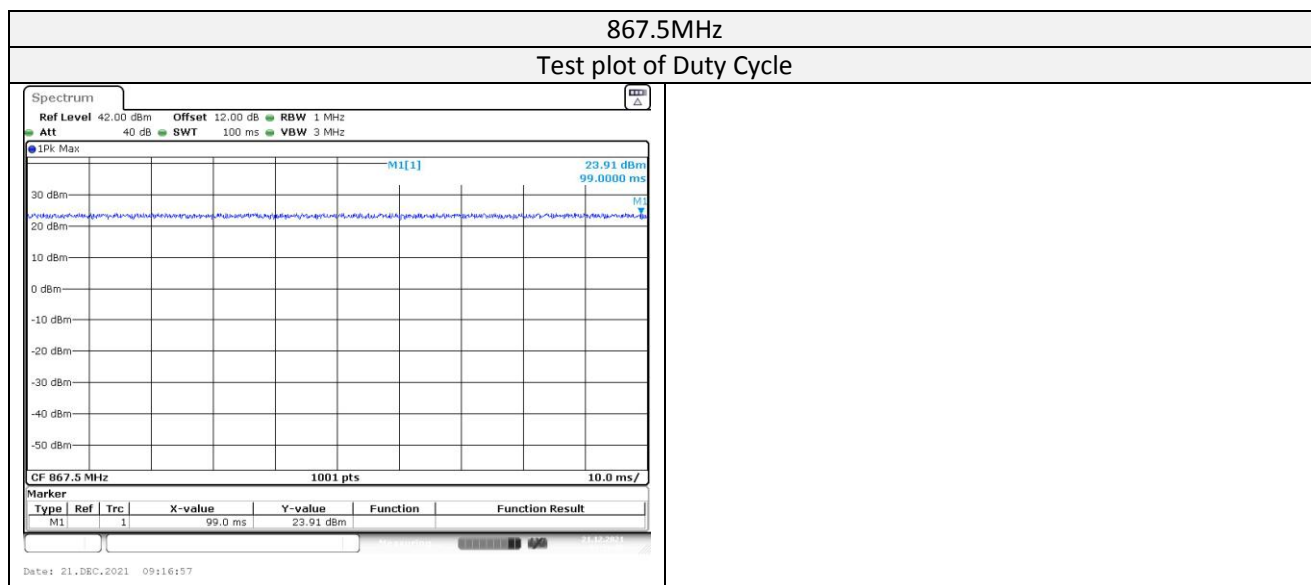
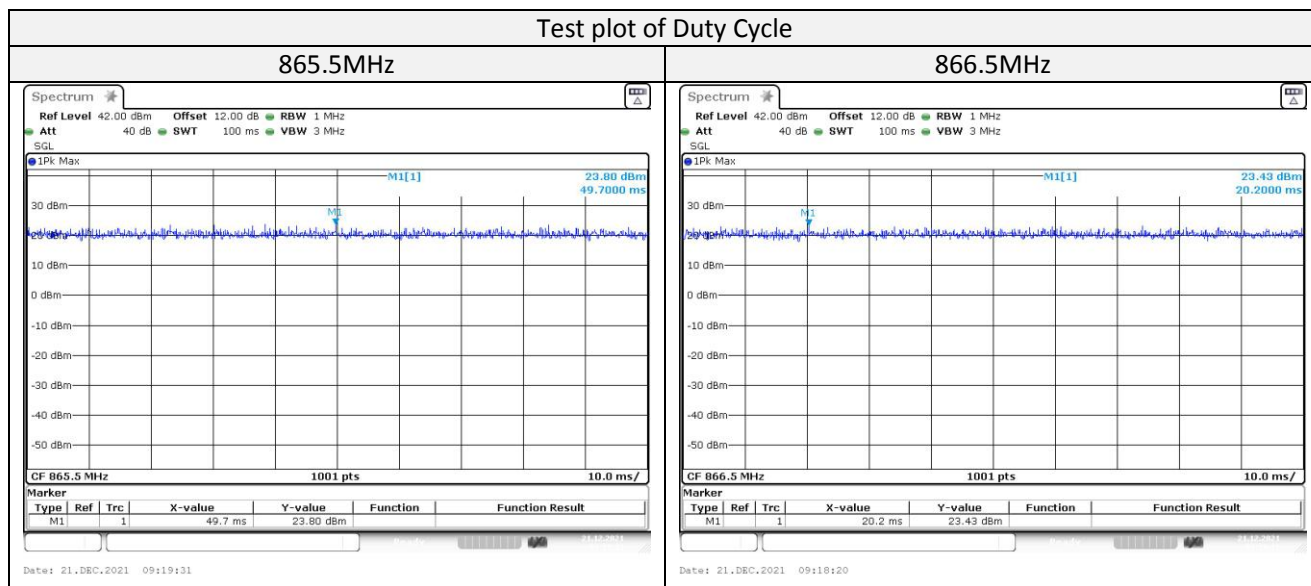
865.0-868.0MHz:

Channel Bandwidth: 1MHz

Frequency (MHz)	Transmit On time (ms)	Duty Cycle	Limit
865.5	100	100%	/
866.5	100	100%	/
867.5	100	100%	/

Note: The duty cycle is calculated as follow:

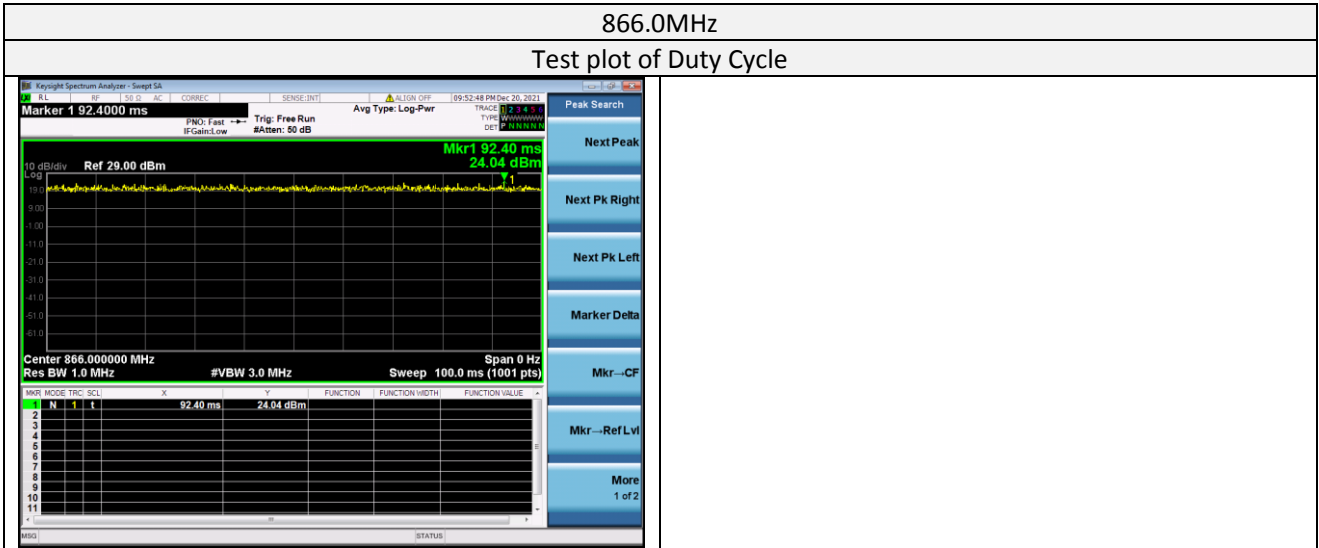
$$\text{Duty cycle} = \frac{\text{Transmit On time (s)}(\text{Within 1 hour})}{3600(\text{s})} * 100\%$$



Channel Bandwidth: 2MHz

Frequency (MHz)	Transmit On time (ms)	Duty Cycle	Limit
866.0	100	100%	/

Note: The duty cycle is calculated as follow:
Duty cycle= Transmit On time (s)(Within 1 hour)/3600(s) *100%



Note:EUT is a Polite spectrum access device.

7. OPERATING FREQUENCY & OCCUPIED BANDWIDTH

7.1. Standard Applicable

According to ETSI EN 300 220-2 V3.1.1 (2017-02) section 4.2.1 & 4.3.4, the Operating channel(s) shall be entirely within operational frequency bands allowed by annexes B, C or any NRI.

Operational Frequency Band		Maximum effective radiated power, e.r.p.	Maximum occupied bandwidth	Channel access and occupation rules (e.g. Duty cycle or LBT + AFA)
K	863 MHz to 865 MHz	25 mW	The whole band except for audio & video applications limited to 300 kHz	Polite spectrum access
L	865 MHz to 868 MHz	25 mW	The whole band	Polite spectrum access

7.2. Test Procedure

- 1) The measurement shall be performed on the lowest and the highest Operating Frequencies declared by the manufacturer. Additional frequencies may be tested.
- 2) The measurement shall be performed with a spectrum analyser.
- 3) For devices with e.r.p. ≤ -30 dBm, OBW may be either measured or taken as equal to the OCW within the operational frequency band.

Radiated measurement:

A suitable test site shall be selected from those described in clause C.1 and the measurements in clause 5.6.3.4 shall be performed using corresponding radiated measurement methods described in clause C.5.

Conducted measurement:

The EUT shall be connected to an artificial antenna which shall be connect to the test equipment via an appropriate attenuator.

The spectrum analyser shall be configured as appropriate for the parameters shown in the following Table.

Setting	Value	Notes
Centre frequency	The nominal Operating Frequency	The highest or lowest Operating Frequency as declared by the manufacturer
RBW	1 % to 3 % of OCW without being below 100 Hz	/
VBW	3*RBW	Nearest available analyser setting to 3 x RBW
Span	At least 2 x Operating Channel width	Span should be large enough to include all major components of the signal and its side bands
Detector Mode	RMS	
Trace	Max hold	

If the equipment is capable of producing an unmodulated carrier and the test in ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.7 is performed, then the OBW measurements need only be performed under normal test conditions. Any required results for Maximum OBW under extreme conditions are obtained by addition and subtraction of the upper and lower frequency error results to each bandwidth measurement obtained in this test.

Step 1:

Operation of the EUT shall be started, on the highest operating frequency as declared by the manufacturer, with the appropriate test signal.

The signal attenuation shall be adjusted to ensure that the signal power envelope is sufficiently above the noise floor of the analyser to avoid the noise signals on either side of the power envelope being included in the measurement.

Step 2:

When the trace is completed the peak value of the trace shall be located and the analyser marker placed on this peak.

Step 3:

The 99 % occupied bandwidth function of the spectrum analyser shall be used to measure the occupied bandwidth of the signal.

7.3. Test Data**Environmental Conditions**

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 KPa

Test Mode: Transmitting

863.0-865.0MHz:

Channel Bandwidth: 1MHz(863.5MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	863.03038	863.95931	0.92892	$865 \leq f \leq 868$
	DC 5.0V	863.03047	863.95954	0.92907	$865 \leq f \leq 868$
	DC 5.5V	863.03044	863.95946	0.92903	$865 \leq f \leq 868$
25	DC 4.5V	863.03042	863.95937	0.92894	$865 \leq f \leq 868$
	DC 5.0V	863.03046	863.95946	0.92900	$865 \leq f \leq 868$
	DC 5.5V	863.03043	863.95951	0.92908	$865 \leq f \leq 868$
45	DC 4.5V	863.03048	863.95948	0.92900	$865 \leq f \leq 868$
	DC 5.0V	863.03035	863.95936	0.92900	$865 \leq f \leq 868$
	DC 5.5V	863.03041	863.95938	0.92897	$865 \leq f \leq 868$

Channel Bandwidth: 1MHz(864.5MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	864.03553	864.96453	0.92900	$865 \leq f \leq 868$
	DC 5.0V	864.03546	864.96454	0.92907	$865 \leq f \leq 868$
	DC 5.5V	864.03545	864.96436	0.92891	$865 \leq f \leq 868$
25	DC 4.5V	864.03555	864.96455	0.92900	$865 \leq f \leq 868$
	DC 5.0V	864.03554	864.96448	0.92894	$865 \leq f \leq 868$
	DC 5.5V	864.03546	864.96450	0.92903	$865 \leq f \leq 868$
45	DC 4.5V	864.03535	864.96426	0.92890	$865 \leq f \leq 868$
	DC 5.0V	864.03536	864.96427	0.92892	$865 \leq f \leq 868$
	DC 5.5V	864.03545	864.96438	0.92893	$865 \leq f \leq 868$

Channel Bandwidth: 2MHz(864.0MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	863.05929	864.94049	1.88120	$865 \leq f \leq 868$
	DC 5.0V	863.05930	864.94070	1.88133	$865 \leq f \leq 868$
	DC 5.5V	863.05925	864.94045	1.88120	$865 \leq f \leq 868$
25	DC 4.5V	863.05936	864.94051	1.88115	$865 \leq f \leq 868$
	DC 5.0V	863.05922	864.94037	1.88115	$865 \leq f \leq 868$
	DC 5.5V	863.05921	864.94035	1.88114	$865 \leq f \leq 868$
45	DC 4.5V	863.05937	864.94055	1.88118	$865 \leq f \leq 868$
	DC 5.0V	863.05931	864.94043	1.88113	$865 \leq f \leq 868$
	DC 5.5V	863.05925	864.94055	1.88130	$865 \leq f \leq 868$

865.0-868.0MHz:

Channel Bandwidth: 1MHz(865.5MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	865.03498	865.95839	0.92341	$865 \leq f \leq 868$
	DC 5.0V	865.03546	865.95954	0.92408	$865 \leq f \leq 868$
	DC 5.5V	865.03502	865.95848	0.92346	$865 \leq f \leq 868$
25	DC 4.5V	865.03509	865.95857	0.92349	$865 \leq f \leq 868$
	DC 5.0V	865.03491	865.95839	0.92348	$865 \leq f \leq 868$
	DC 5.5V	865.03494	865.95837	0.92343	$865 \leq f \leq 868$
45	DC 4.5V	865.03508	865.95859	0.92352	$865 \leq f \leq 868$
	DC 5.0V	865.03497	865.95843	0.92347	$865 \leq f \leq 868$
	DC 5.5V	865.03494	865.95838	0.92344	$865 \leq f \leq 868$

Channel Bandwidth: 1MHz(866.5MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	866.04005	866.95848	0.91843	$865 \leq f \leq 868$
	DC 5.0V	866.04046	866.95954	0.91908	$865 \leq f \leq 868$
	DC 5.5V	866.03991	866.95836	0.91845	$865 \leq f \leq 868$
25	DC 4.5V	866.03999	866.95844	0.91844	$865 \leq f \leq 868$
	DC 5.0V	866.03995	866.95846	0.91851	$865 \leq f \leq 868$
	DC 5.5V	866.04004	866.95845	0.91841	$865 \leq f \leq 868$
45	DC 4.5V	866.03990	866.95846	0.91856	$865 \leq f \leq 868$
	DC 5.0V	866.04004	866.95853	0.91849	$865 \leq f \leq 868$
	DC 5.5V	866.04009	866.95855	0.91846	$865 \leq f \leq 868$

Channel Bandwidth: 1MHz(867.5MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	867.02992	867.96340	0.93347	$865 \leq f \leq 868$
	DC 5.0V	867.03047	967.96454	0.93407	$865 \leq f \leq 868$
	DC 5.5V	867.02991	867.96350	0.93359	$865 \leq f \leq 868$
25	DC 4.5V	867.03008	867.96355	0.93346	$865 \leq f \leq 868$
	DC 5.0V	867.03002	867.96347	0.93345	$865 \leq f \leq 868$
	DC 5.5V	867.03007	867.96350	0.93343	$865 \leq f \leq 868$
45	DC 4.5V	867.02991	867.96343	0.93352	$865 \leq f \leq 868$
	DC 5.0V	867.02995	867.96351	0.93355	$865 \leq f \leq 868$
	DC 5.5V	867.02996	867.96347	0.93350	$865 \leq f \leq 868$

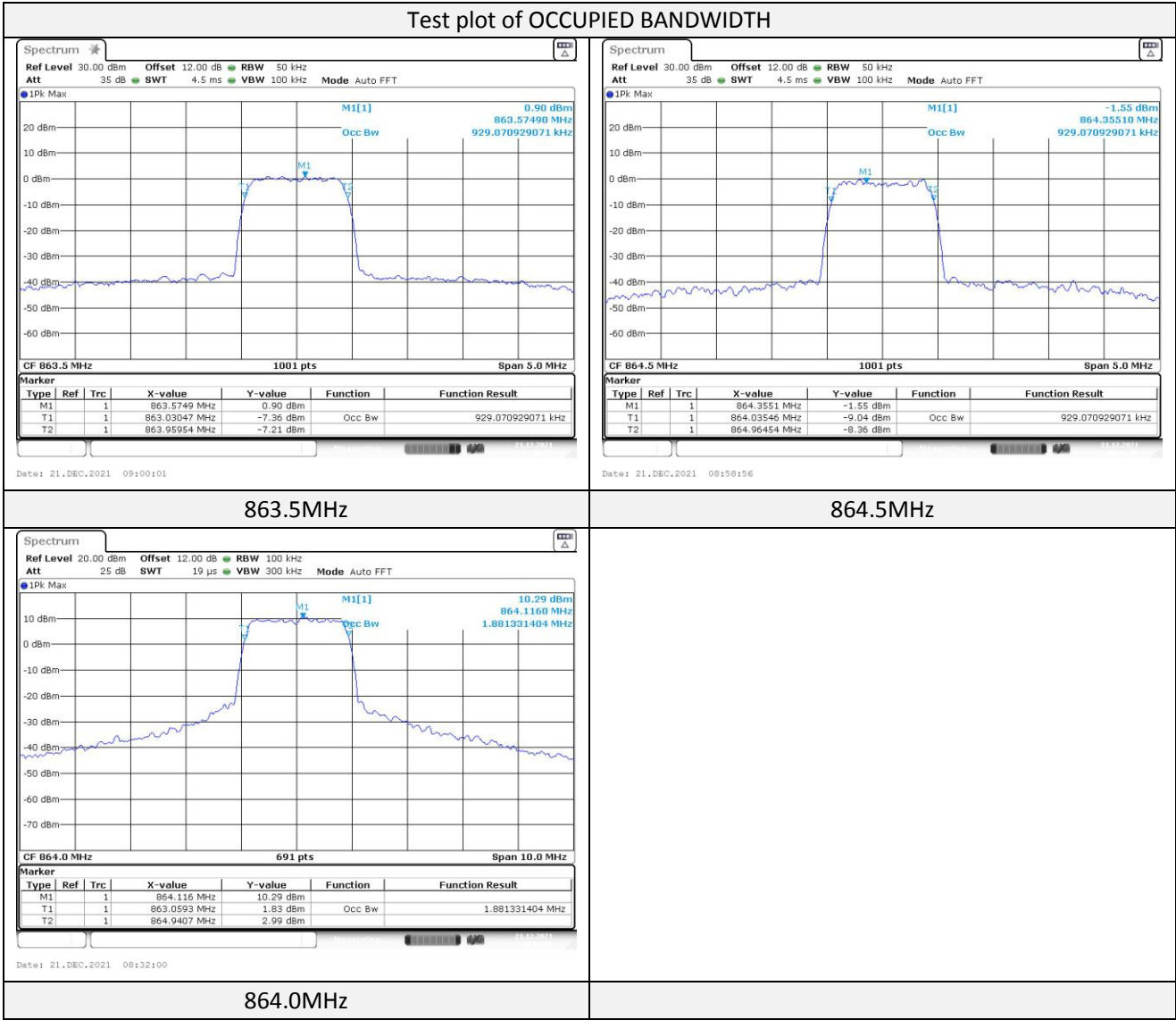
Channel Bandwidth: 2MHz(866.0MHz)

Environment Temperature (°C)	Power Supplied V	F _{lower} (MHz)	F _{upper} (MHz)	OCW(MHz)	Limit(MHz)
-20	DC 4.5V	865.13008	866.89612	1.76604	$865 \leq f \leq 868$
	DC 5.0V	865.13544	866.90384	1.76840	$865 \leq f \leq 868$
	DC 5.5V	865.12999	866.89593	1.76594	$865 \leq f \leq 868$
25	DC 4.5V	865.12992	866.89599	1.76607	$865 \leq f \leq 868$
	DC 5.0V	865.13001	866.89603	1.76603	$865 \leq f \leq 868$
	DC 5.5V	865.12999	866.89604	1.76605	$865 \leq f \leq 868$
45	DC 4.5V	865.12999	866.89597	1.76598	$865 \leq f \leq 868$
	DC 5.0V	865.12995	866.89586	1.76591	$865 \leq f \leq 868$
	DC 5.5V	865.13008	866.89603	1.76595	$865 \leq f \leq 868$

Note: The test signal used. Please refer to the Table 2 in ETSI EN 300 220-1 V3.1.1 (2017-02).

Highest measured OBW value or if the measurement is only performed at normal temperature conditions, the upper and lower frequency error results have to be added and subtracted to measured OBW to calculate the Maximum Occupied Bandwidth.

863.0-865.0MHz:

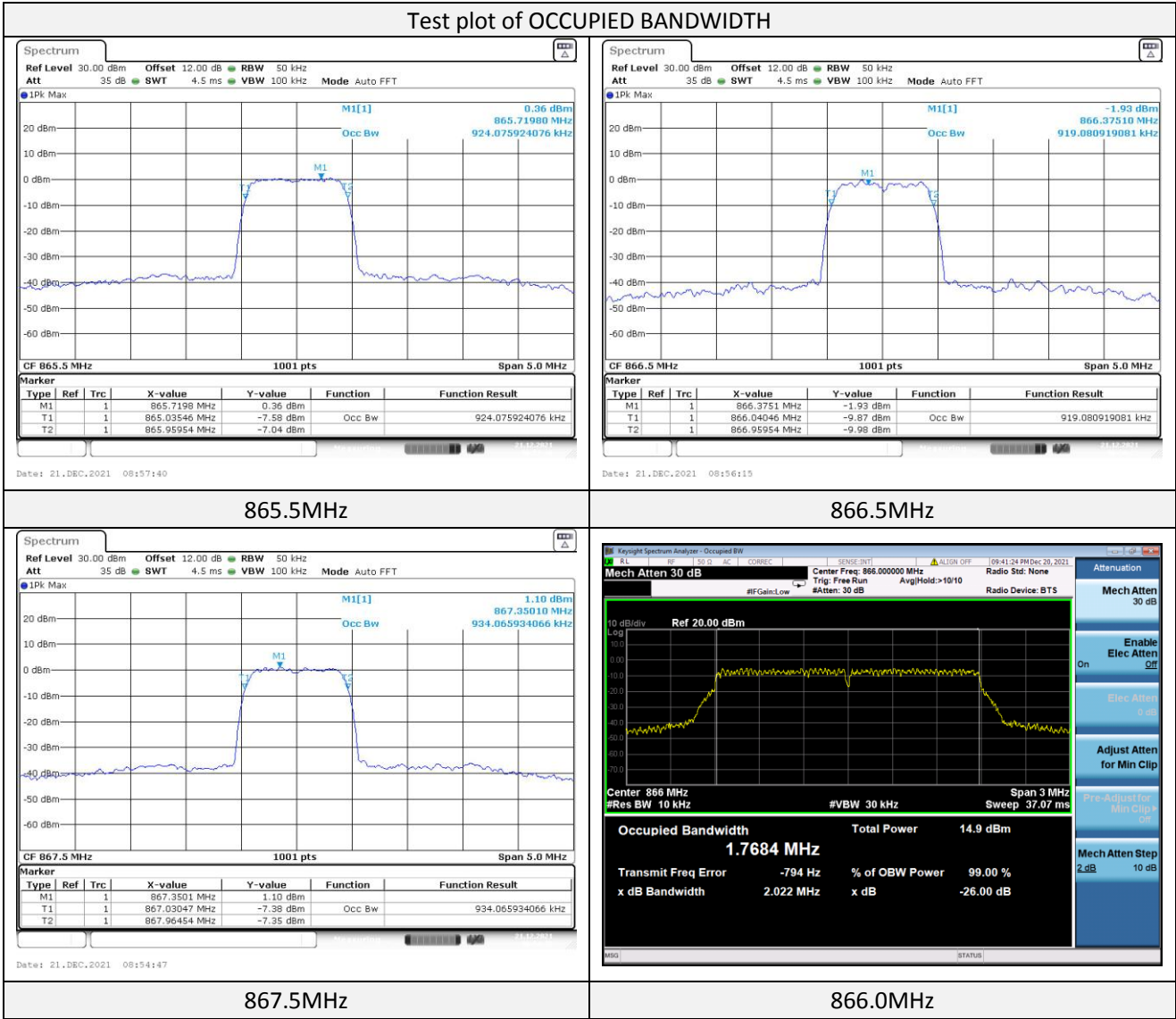


863.5MHz

864.5MHz

864.0MHz

865.0-868.0MHz:



8. TX OUT OF BAND EMISSIONS

8.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.8.

Two OOB domains are defined, one for OC (see Figure 2) and one for Operational Frequency band (see Figure 3).

The spectrum masks for these two OOB domains may overlap.

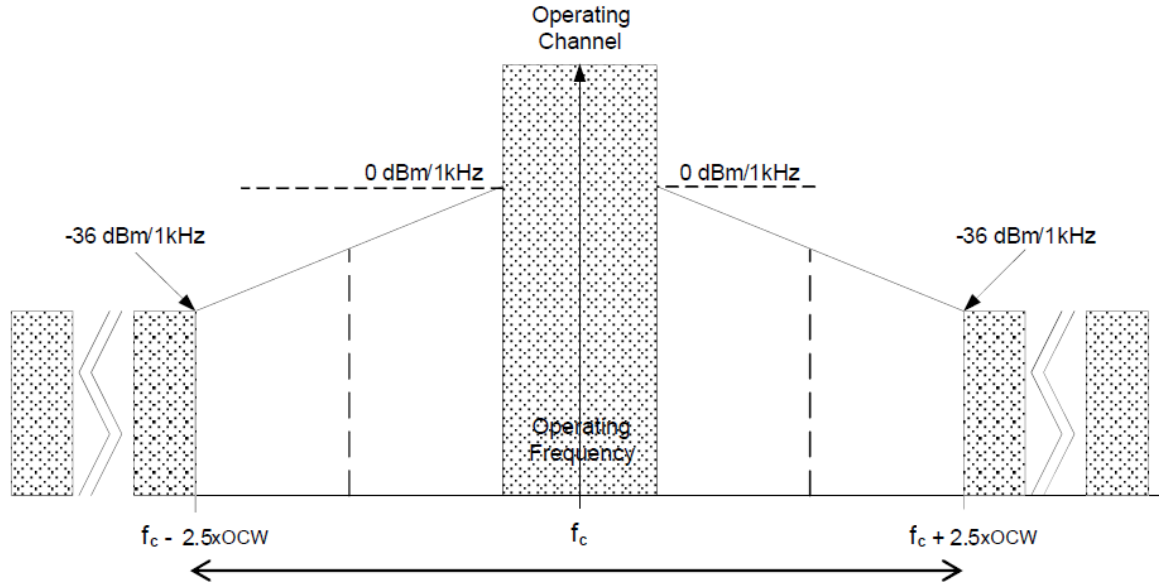


Figure 2: Out Of Band Domain for Operating Channel with reference BW

Unwanted emissions in the Out Of Band domain are those falling in the frequency range immediately below the lower, and above the upper, frequency of the Operating Channel. The OOB domain includes both frequencies outside the Operating Channel within the Operational Frequency Band and frequencies outside the Operational Frequency Band.

The relevant Out Of Band domain is shown in Figure 2 and applies within the Operational Frequency Band.

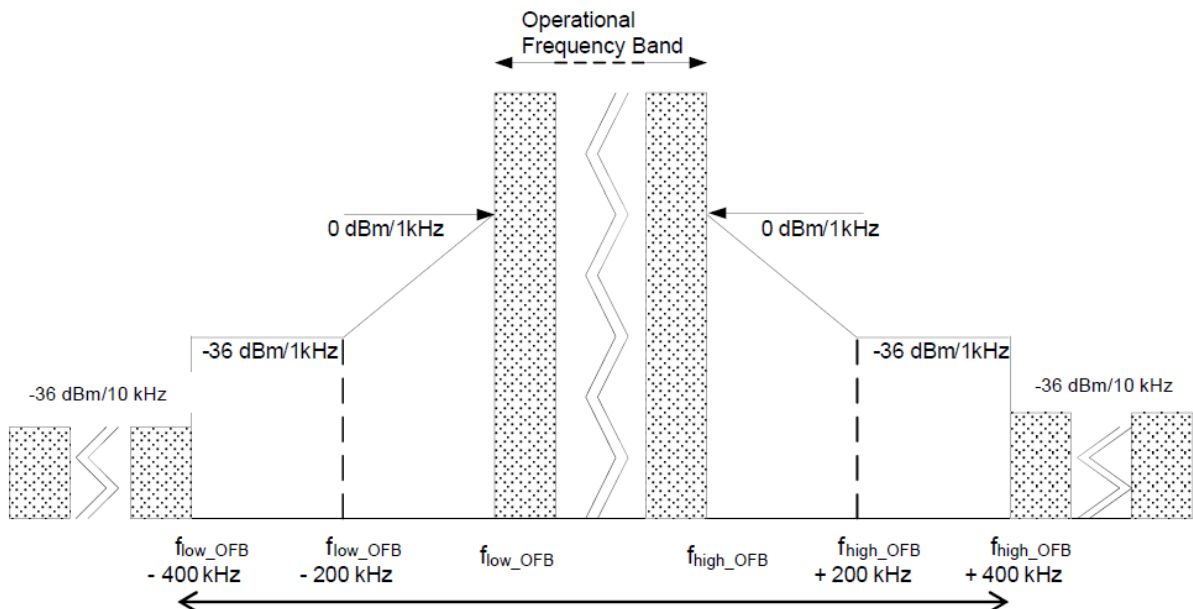


Figure 3: Out Of Band Domain for Operational Frequency Band with reference BW

Specific limits apply at frequencies immediately above and below the Operational Frequency Band as shown in Figure 3.

NOTE: f_{low_OFB} is the lower edge of the Operational Frequency Band.

f_{high_OFB} is the upper edge of the Operational Frequency Band.

The EUT emissions level in OOB domains for the Operating Channel and the Operational Frequency Band shall be less or equal to Table 1 spectrum mask.

Table 1: Emission limits in the Out Of Band domains

Domain	Frequency Range	RBW _{REF}	Max power limit
OOB limits applicable to Operational Frequency Band (See Figure 3)	$f \leq f_{\text{low_OFB}} - 400 \text{ kHz}$	10KHz	-36dBm
	$F_{\text{low_OFB}} - 400 \text{ kHz} \leq f \leq f_{\text{low_OFB}} - 200 \text{ kHz}$	1KHz	-36 dBm
	$f_{\text{low}} - 200 \text{ kHz} \leq f < f_{\text{low_OFB}}$	1KHz	See Figure 3
	$f = f_{\text{low_OFB}}$	1KHz	0 dBm
	$f = f_{\text{high_OFB}}$	1KHz	0 dBm
	$F_{\text{high_OFB}} < f \leq f_{\text{high_OFB}} + 200 \text{ kHz}$	1KHz	See Figure 3
	$F_{\text{high_OFB}} + 200 \text{ kHz} \leq f \leq f_{\text{high_OFB}} + 400 \text{ kHz}$	1KHz	-36 dBm
	$F_{\text{high_OFB}} + 400 \text{ kHz} \leq f$	10KHz	-36 dBm
OOB limits applicable to Operating Channel (See Figure 2)	$f = f_c - 2.5 \times \text{OCW}$	1KHz	-36 dBm
	$f_c - 2.5 \times \text{OCW} \leq f \leq f_c - 0.5 \times \text{OCW}$	1KHz	See Figure 2
	$f = f_c - 0.5 \times \text{OCW}$	1KHz	0 dBm
	$f = f_c + 0.5 \times \text{OCW}$	1KHz	0 dBm
	$f_c + 0.5 \times \text{OCW} \leq f \leq f_c + 2.5 \times \text{OCW}$	1KHz	See Figure 2
	$f = f_c + 2.5 \times \text{OCW}$	1KHz	-36 dBm
NOTE: f is the measurement frequency. f_c is the Operating Frequency. $F_{\text{low_OFB}}$ is the lower edge of the Operational Frequency Band. $F_{\text{high_OFB}}$ is the upper edge of the Operational Frequency Band. OCW is the operating channel bandwidth.			

8.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.8.3 for the measurement method.

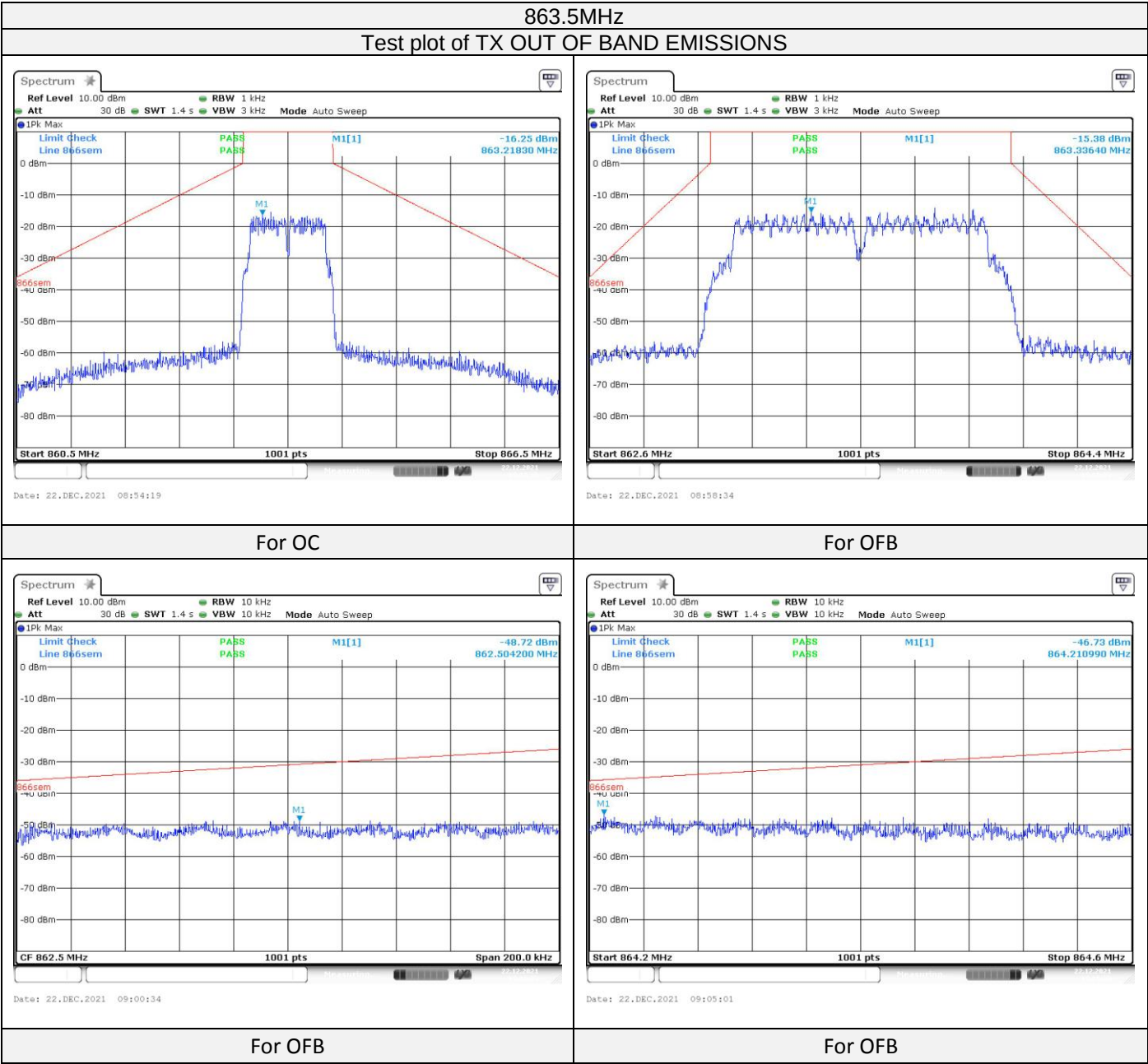
8.3. Test Data

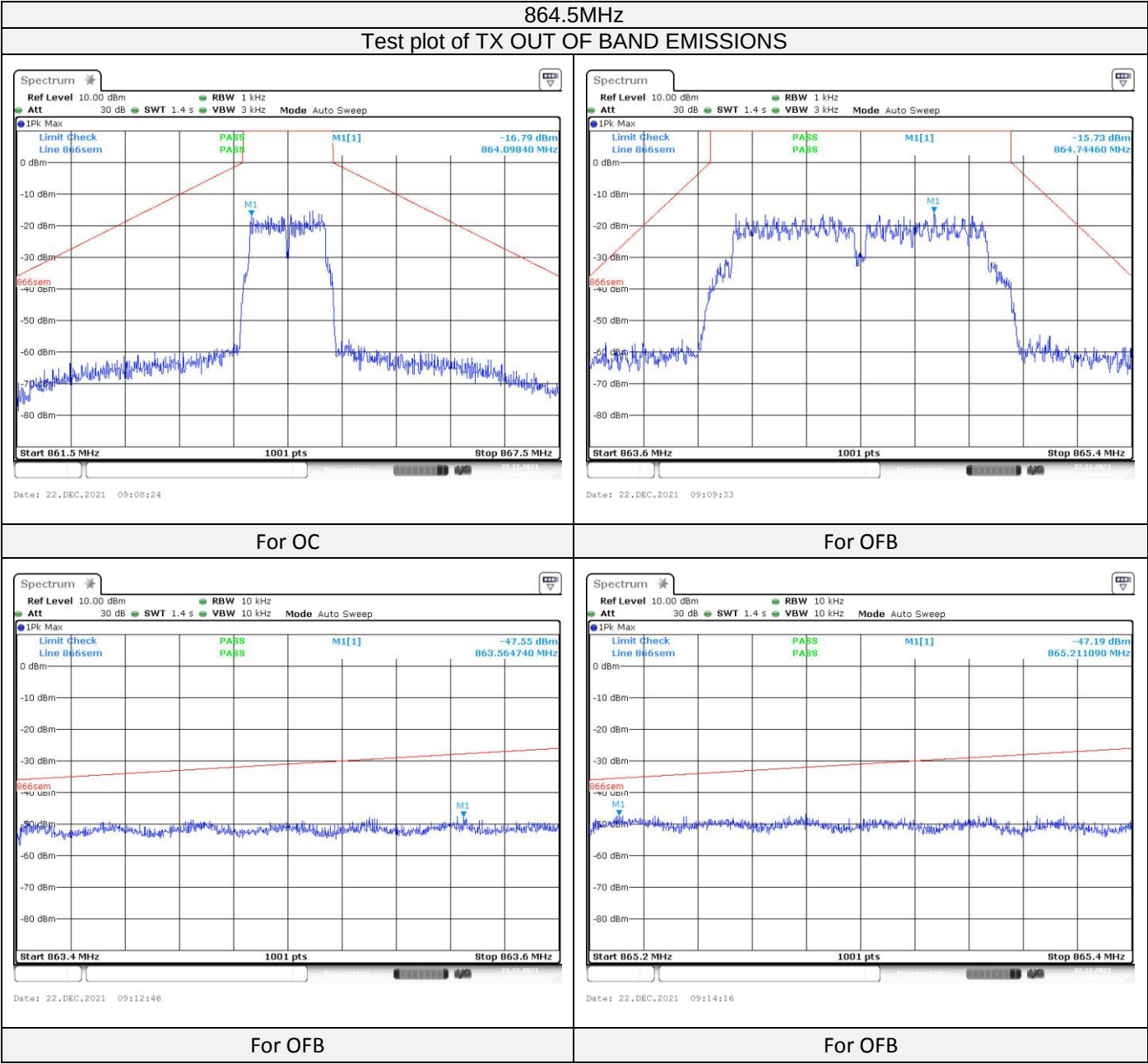
Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

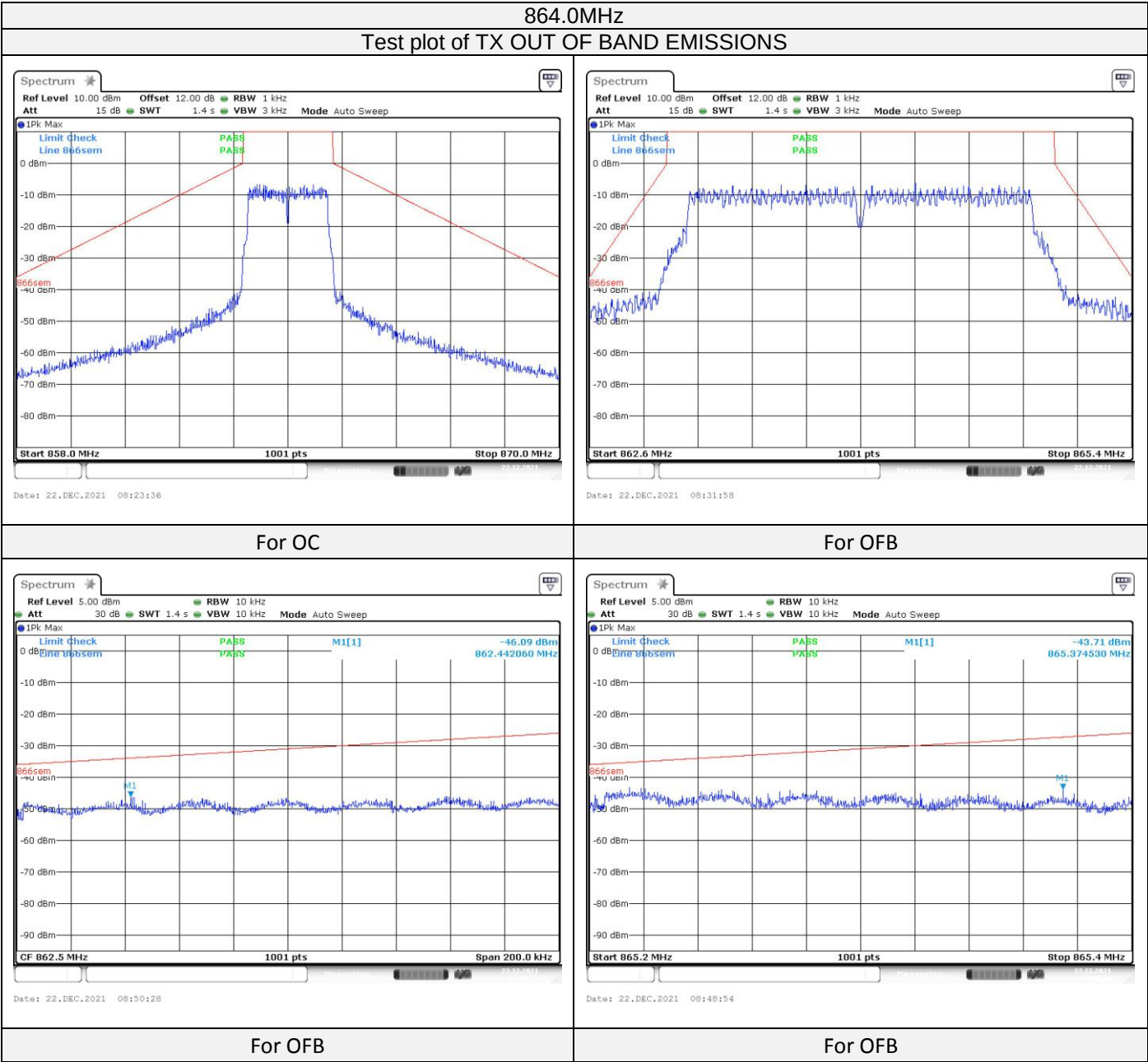
863.0-865.0MHz:

Channel Bandwidth: 1MHz



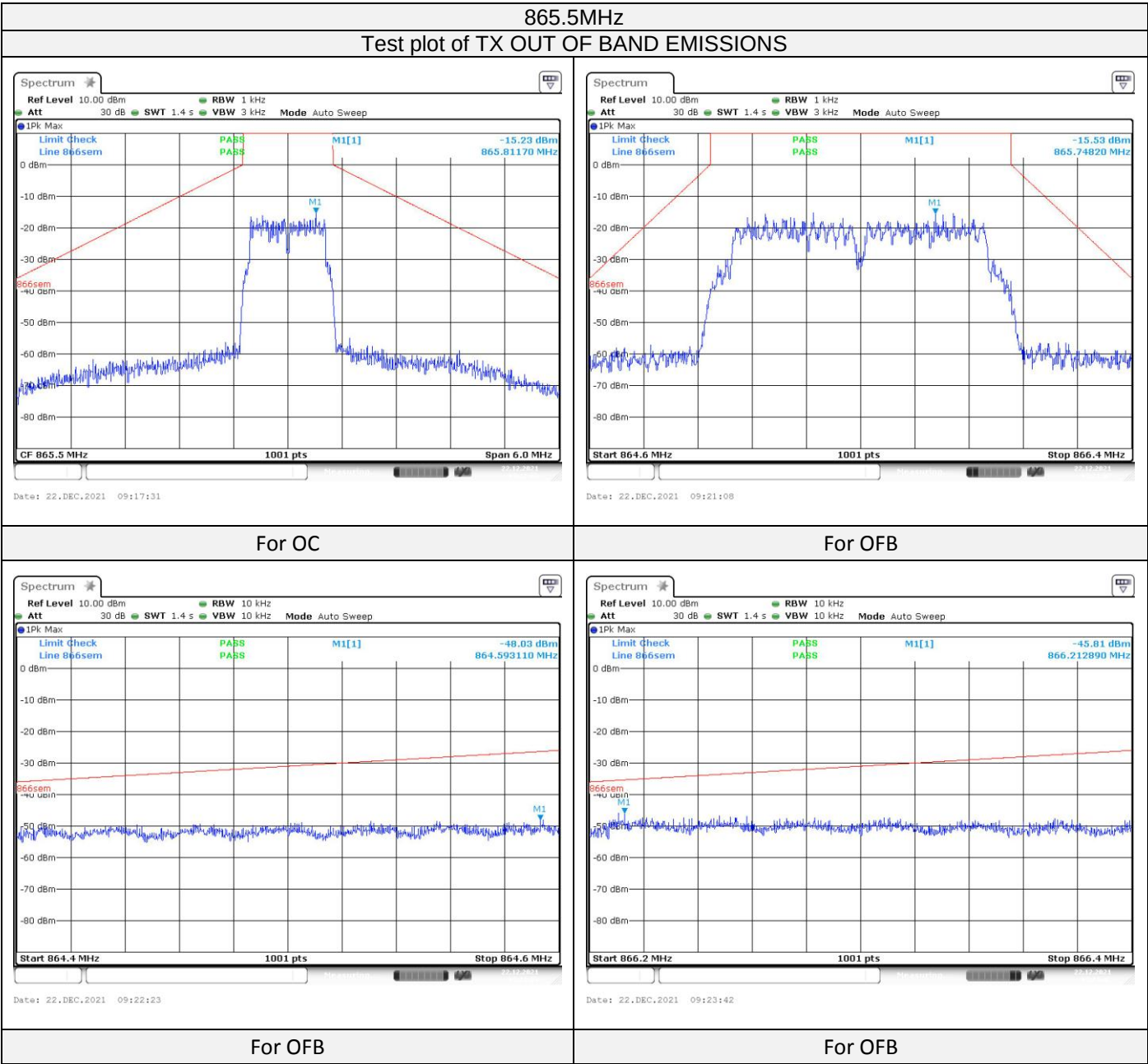


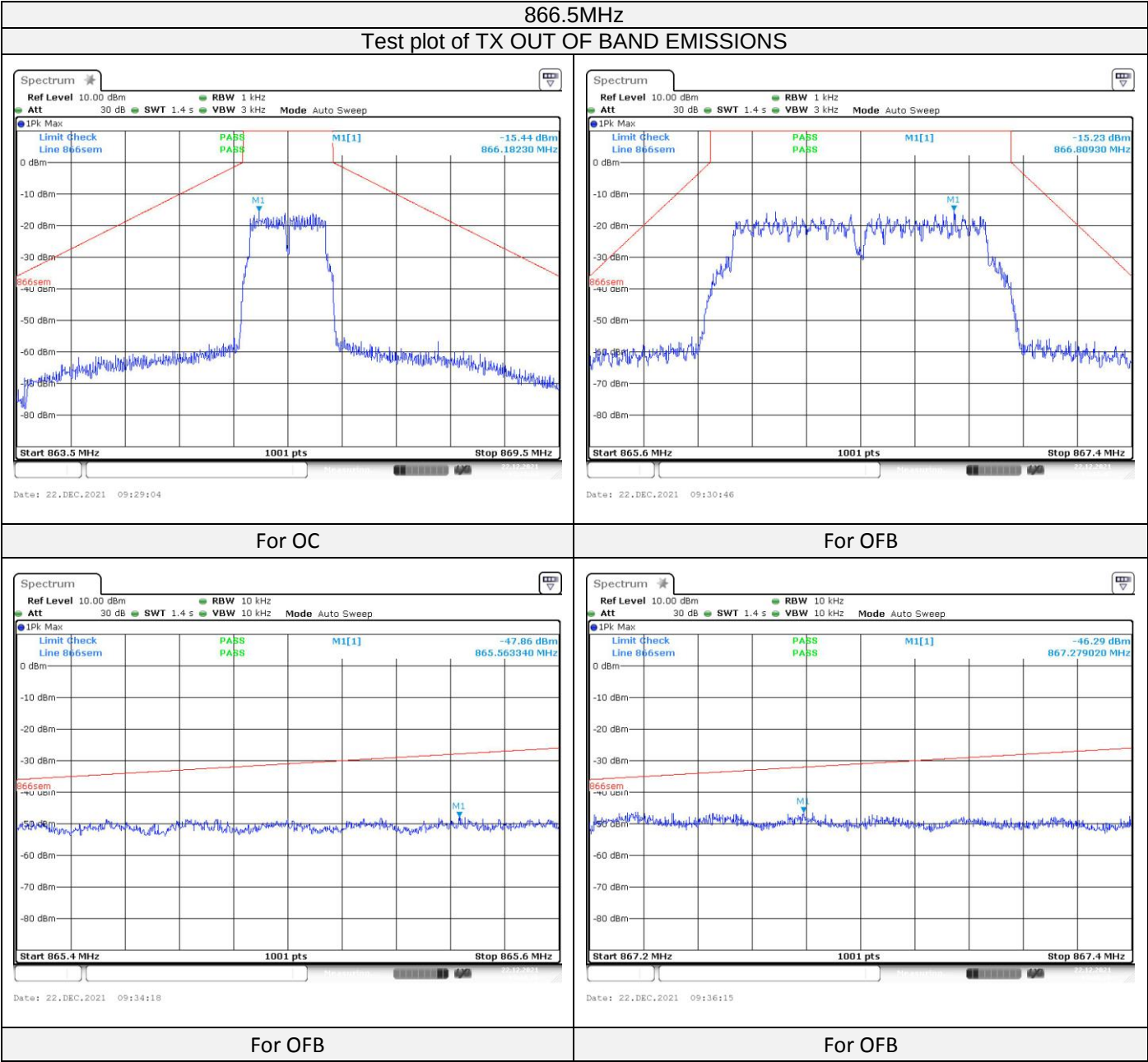
Channel Bandwidth: 2MHz

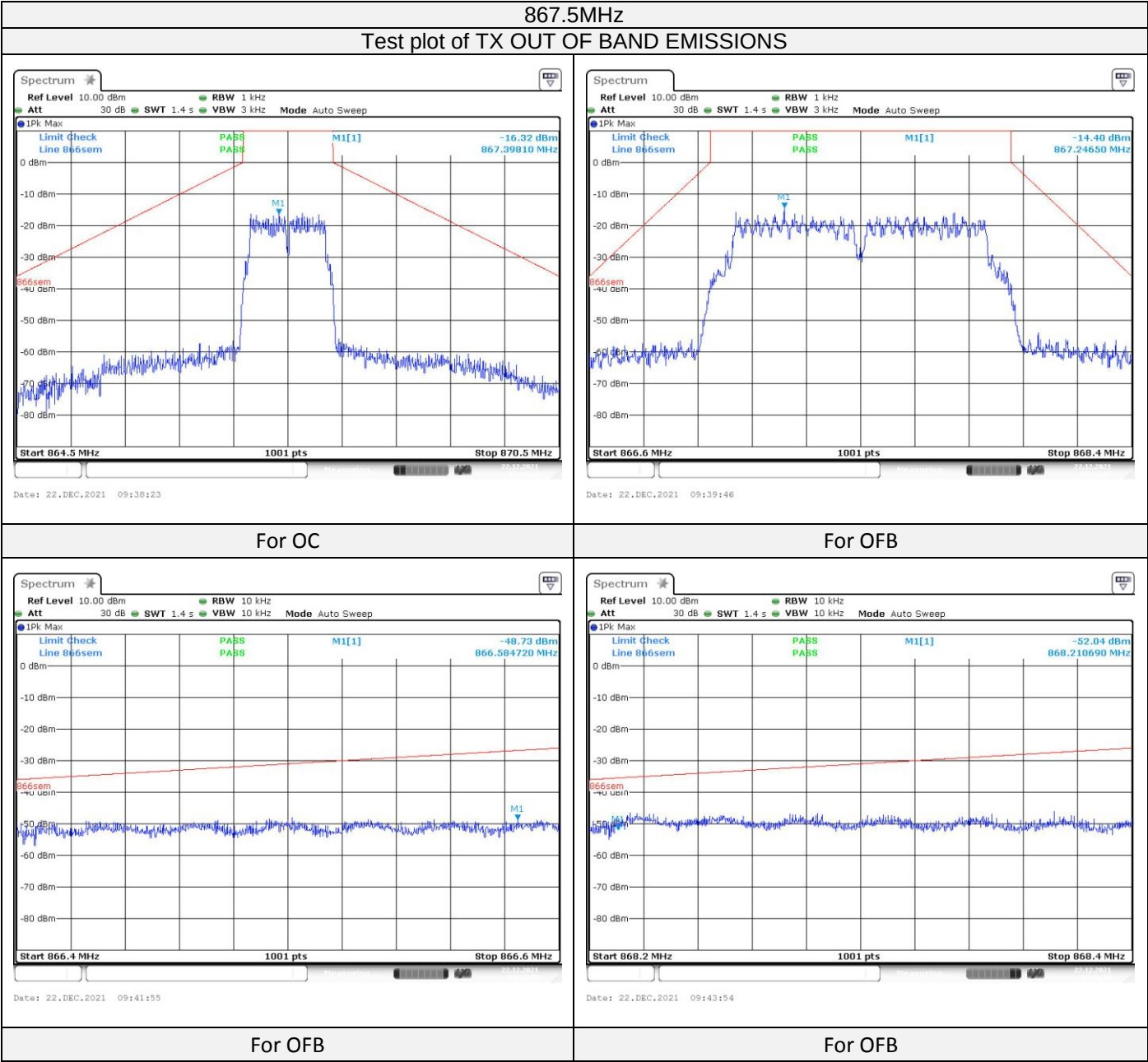


865.0-868.0MHz:

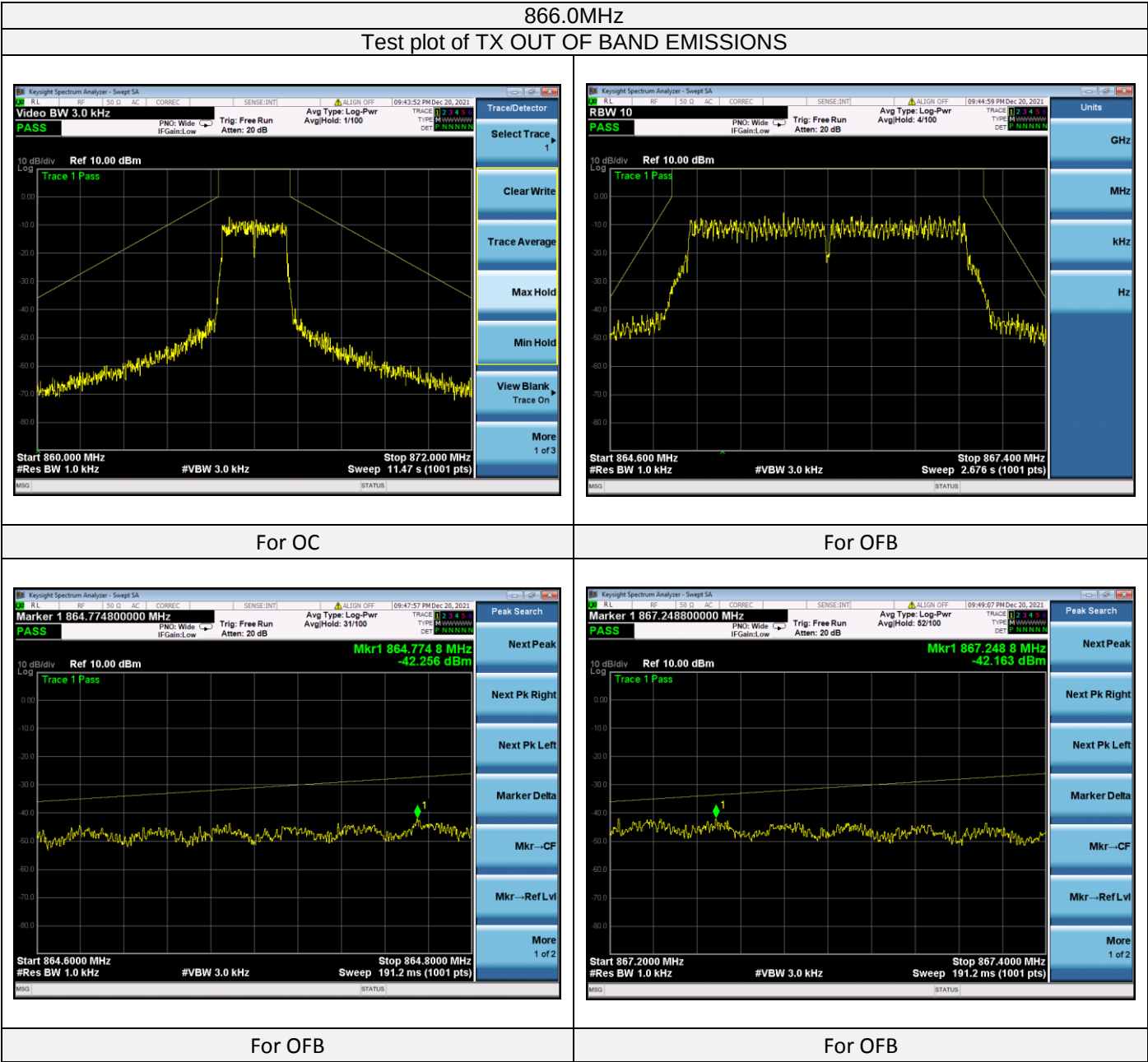
Channel Bandwidth: 1MHz







Channel Bandwidth: 2MHz



Note: 1. OC is the operating channel;
2. OFB is the Operational Frequency Band.

9. TRANSIENT POWER

9.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.10.

Transmitter transient power is power falling into frequencies other than the operating channel as a result of the transmitter being switched on and off.

The transient power shall not exceed the values given in Table 4.

Table 4: Transmitter Transient Power limits

Absolute offset from centre frequency	RBW _{REF}	Peak power limit applicable at measurement points
≤ 400 kHz	1KHz	0dBm
> 400 kHz	1KHz	-27dBm

9.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.10.3 for the measurement method.

9.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Mode: Transmitting

863.0-865.0MHz:

Channel Bandwidth: 1MHz(863.5MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-47.75	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-43.94	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-44.69	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-44.20	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-54.55	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-54.32	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-56.55	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-57.62	-27	

Channel Bandwidth: 1MHz(864.5MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-44.14	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-42.38	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-45.98	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-44.85	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-52.29	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-55.74	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-55.06	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-55.12	-27	

Channel Bandwidth: 2MHz(864.0MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-45.75	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-43.97	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-44.09	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-42.48	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-54.68	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-53.65	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-55.33	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-53.67	-27	

865.0-868.0MHz:

Channel Bandwidth: 1MHz(865.5MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-47.06	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-44.58	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-47.65	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-46.30	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-53.17	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-53.56	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-58.36	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-53.12	-27	

Channel Bandwidth: 1MHz(866.5MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-47.01	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-44.30	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-47.63	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-45.78	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-52.38	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-52.67	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-58.38	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-55.02	-27	

Channel Bandwidth: 1MHz(867.5MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-46.10	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-45.62	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-47.76	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-46.87	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-55.08	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-55.84	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-57.18	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-56.60	-27	

Channel Bandwidth: 2MHz(866.0MHz)

Test condition			Transient power(dBm)	Limit(dBm)	Text result(pass/fail)
Step 1	Upper frequency	Fc+0,5 x OCW + 3 kHz	-46.75	-27	PASS
	Lower frequency	Fc-0,5 x OCW - 3 kHz	-42.36	-27	
Step 2	Upper Frequency	Fc+12,5 kHz or fc+OCW whichever is the greater	-44.28	-27	
	Lower Frequency	Fc-12,5 kHz or fc-OCW whichever is the greater	-45.65	-27	
Step 3	Upper Frequency	Fc+0,5xOCW+400 kHz	-53.18	-27	
	Lower Frequency	Fc-0,5xOCW-400 kHz	-52.45	-27	
Step 4	Upper Frequency	Fc+0,5xOCW+1200 kHz	-56.18	-27	
	Lower Frequency	Fc-0,5xOCW-1200 kHz	-55.83	-27	

10. ADJACENT CHANNEL POWER

10.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.11.

Adjacent channel power is power incidental to proper operation of a transmitter falling into the neighbouring channels.

Where the operating channel width is less than or equal to 25 kHz, the power in the adjacent channels shall not exceed the values given in Table 5.

Table 5: Adjacent channel power limits for transmitters with OCW $\leq$ 25 kHz

		Adjacent Channel power integrated over 0,7 x OCW	Alternate Adjacent Channel power integrated over 0,7 x OCW
OCW < 20 kHz	Normal test conditions	-20 dBm	-20 dBm
	Extreme test conditions	-15 dBm	-20 dBm
OCW $\geq$ 20 kHz	Normal test conditions	-37 dBm	-40 dBm
	Extreme test conditions	-32 dBm	-37 dBm

10.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.11.3 for the measurement method.

10.3. Test Data

Not applicable: EUT No requirement.

Because the OCW of this EUT was great than 25 KHz.

11. TX BEHAVIOUR UNDER LOW VOLTAGE CONDITIONS

11.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.12.

The TX behaviour under low voltage condition is the ability of the equipment to maintain its operating frequency and not produce emissions which exceed any relevant limit when the battery voltage falls below the lower extreme voltage level.

The equipment shall either:

- a) remain in the Operating Channel OC without exceeding any applicable limits (e.g. Duty Cycle); or
 - b) reduce its effective radiated power below the Spurious Emission limits without exceeding any applicable limits (e.g. Duty Cycle); or
 - c) shut down, (ceasing function); as the voltage falls below the manufacturers declared operating voltage.
- as the voltage falls below the manufacturers declared operating voltage.

11.2. Test Procedure

Step 1:

Operation of the EUT shall be started, on Operating Frequency as declared by the manufacturer, with the appropriate test signal and with the EUT operating at nominal operating voltage.

The centre frequency of the transmitted signal shall be measured and noted.

Step 2:

The operating voltage shall be reduced by appropriate steps until the voltage reaches zero.

The centre frequency of the transmitted signal shall be measured and noted.

Any abnormal behaviour shall be noted.

11.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Mode: Transmit

863.0-865.0MHz:

Channel Bandwidth: 1MHz(863.5MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	863.5	<-36	100%	/	Pass
4.5	863.5	<-36	100%	/	Pass
3.0	863.5	<-36	100%	/	Pass
0	863.5	0	0	/	Pass

Channel Bandwidth: 1MHz(864.5MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	864.5	<-36	100%	/	Pass
4.5	864.5	<-36	100%	/	Pass
3.0	864.5	<-36	100%	/	Pass
0	864.5	0	0	/	Pass

Channel Bandwidth: 2MHz(864.0MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	864.0	<-36	100%	/	Pass
4.5	864.0	<-36	100%	/	Pass
3.0	864.0	<-36	100%	/	Pass
0	864.0	0	0	/	Pass

865.0-868.0MHz:

Channel Bandwidth: 1MHz(865.5MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	865.5	<-36	100%	/	Pass
4.5	865.5	<-36	100%	/	Pass
3.0	865.5	<-36	100%	/	Pass
0	865.5	0	0	/	Pass

Channel Bandwidth: 1MHz(866.5MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	866.5	<-36	100%	/	Pass
4.5	866.5	<-36	100%	/	Pass
3.0	866.5	<-36	100%	/	Pass
0	866.5	0	0	/	Pass

Channel Bandwidth: 1MHz(867.5MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	867.5	<-36	100%	/	Pass
4.5	867.5	<-36	100%	/	Pass
3.0	867.5	<-36	100%	/	Pass
0	867.5	0	0	/	Pass

Channel Bandwidth: 2MHz(866.0MHz)

Voltage Supply (Vdc)	Measurement Frequency (MHz)	ERP (dBm)	Duty Cycle	Limit	Result
5.0	866.0	<-36	100%	/	Pass
4.5	866.0	<-36	100%	/	Pass
3.0	866.0	<-36	100%	/	Pass
0	866.0	0	0	/	Pass

12. ADAPTIVE POWER CONTROL

12.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.13.

The peak power measured when active APC function at its minimum setting shall not exceed the value shown in Table 6.

Table 6: APC power limit

Parameter	Limit
Transmitted e.r.p.	+7 dBm

12.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.13.3 for the measurement method.

12.3. Test Data

Not applicable: EUT No requirement.

13. FHSS EQUIPMENT

13.1. Standard Applicable

According to ETSI EN 300 220-2 V3.1.1 (2017-02) clause 4.3.10.

13.2. Test Procedure

Please refer to ETSI EN 300 220-2 V3.1.1 (2017-02) clause 4.3.10 for the measurement method.

13.3. Test Data

Not applicable: EUT No requirement.

14. SHORT TERM BEHAVIOUR

14.1. Standard Applicable

According to ETSI EN 300 220-2 V3.1.1 (2017-02) clause 4.3.11.

14.2. Test Procedure

Please refer to ETSI EN 300 220-2 V3.1.1 (2017-02) clause 4.3.11 for the measurement method.

14.3. Test Data

Not applicable: EUT No requirement.

15. UNWANTED EMISSIONS IN THE SPURIOUS DOMAIN

15.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.9.2.
The power of any unwanted emission in the spurious domain shall not exceed the values given in Table 7.

Table 19: Spurious domain emission limits

Frequency State	47 MHz to 74 MHz 87,5 MHz to 118 MHz 174 MHz to 230 MHz 470 MHz to 790 MHz	Other frequencies below 1000 MHz	above 1000 MHz
TX mode	-54 dBm	-36 dBm	-30 dBm
RX and all other modes	-57 dBm	-57 dBm	-47 dBm

15.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.9.3 for the measurement method.

15.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

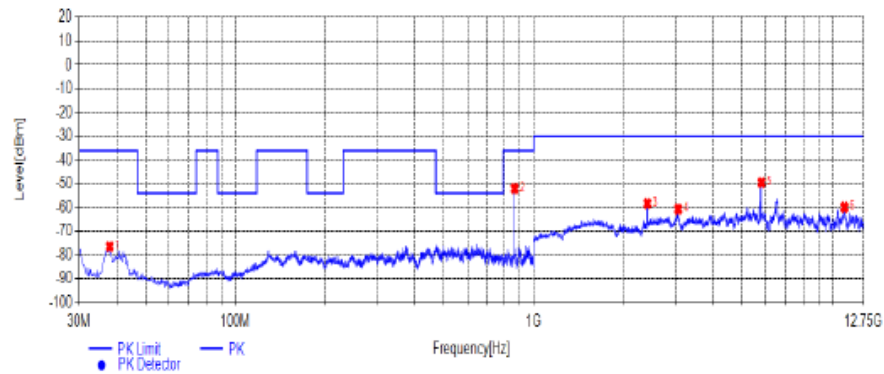
Test Mode: TX

863.0-865.0MHz:

Channel Bandwidth: 1MHz(863.5MHz)

Horizontal

Test Graph



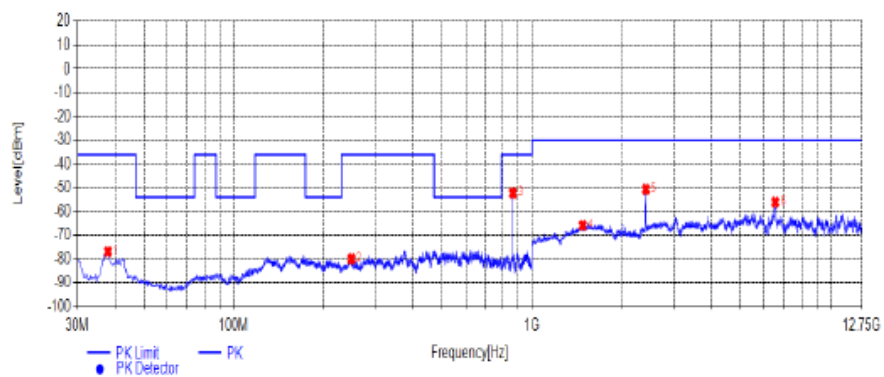
Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.9556	-109.75	40.43	100	250	PK	Horizontal	PASS
2	863.2026	-101.82	16.08	100	320	PK	Horizontal	PASS
3	2407.9316	-99.99	28.30	100	320	PK	Horizontal	PASS
4	3051.9604	-95.31	30.67	100	190	PK	Horizontal	PASS
5	5806.7113	-96.99	19.48	100	280	PK	Horizontal	PASS
6	11001.2503	-88.92	29.94	100	80	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .
 2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

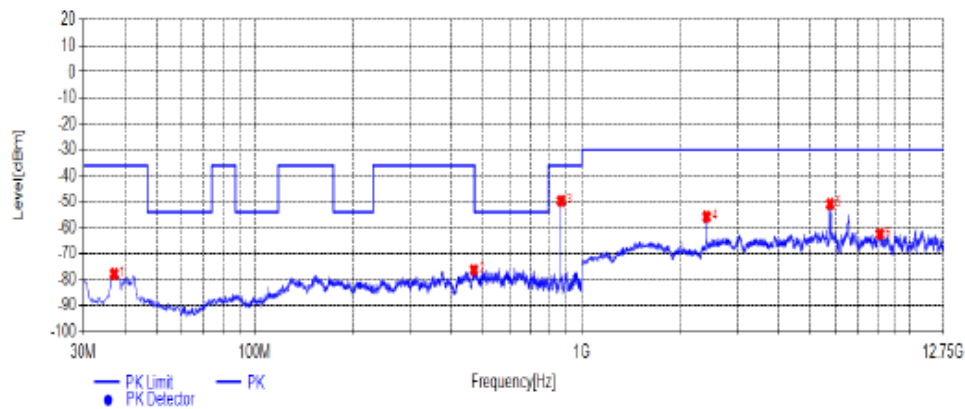
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.9556	-109.75	40.89	100	50	PK	Vertical	PASS
2	248.2937	-103.82	44.04	100	30	PK	Vertical	PASS
3	863.2026	-101.82	16.31	100	90	PK	Vertical	PASS
4	1479.4959	-101.22	35.87	100	40	PK	Vertical	PASS
5	2412.6325	-99.93	20.66	100	330	PK	Vertical	PASS
6	6547.1094	-88.93	26.13	100	90	PK	Vertical	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .
 2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 1MHz(864.5MHz)

Horizontal

Test Graph



Suspected List

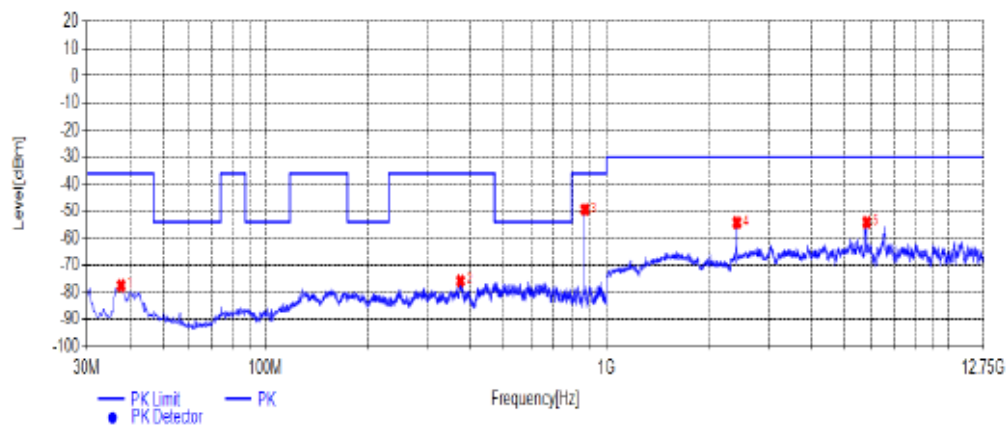
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.1794	-110.33	41.81	100	130	PK	Horizontal	PASS
2	468.3337	-102.11	40.21	100	30	PK	Horizontal	PASS
3	864.3669	-101.89	13.66	100	30	PK	Horizontal	PASS
4	2410.2821	-99.96	25.82	100	350	PK	Horizontal	PASS
5	5750.3001	-96.89	20.88	100	300	PK	Horizontal	PASS
6	8150.1300	-92.91	32.46	100	170	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading (dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.7616	-109.89	41.52	100	60	PK	Vertical	PASS
2	374.0308	-101.13	39.75	100	340	PK	Vertical	PASS
3	864.7550	-101.65	13.36	100	280	PK	Vertical	PASS
4	2414.9830	-99.90	24.21	100	170	PK	Vertical	PASS
5	5799.6599	-97.29	24.09	100	180	PK	Vertical	PASS

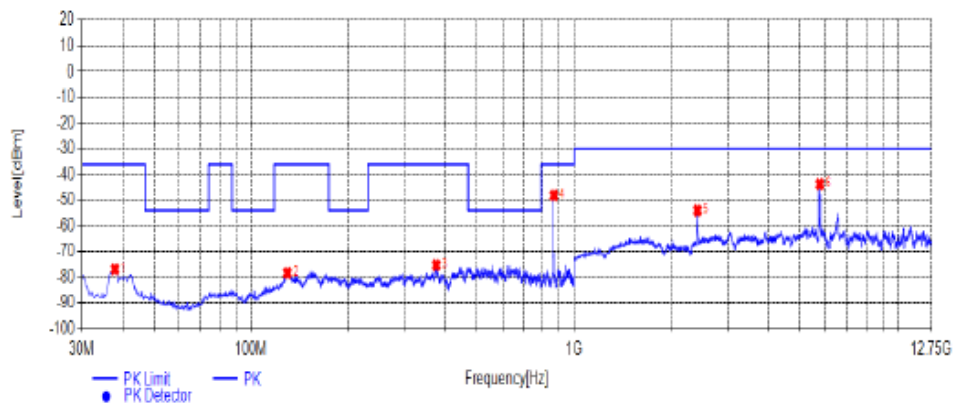
Note:1. Result (dBμV/m) = Reading (dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 2MHz(864.0MHz)

Horizontal

Test Graph



Suspected List

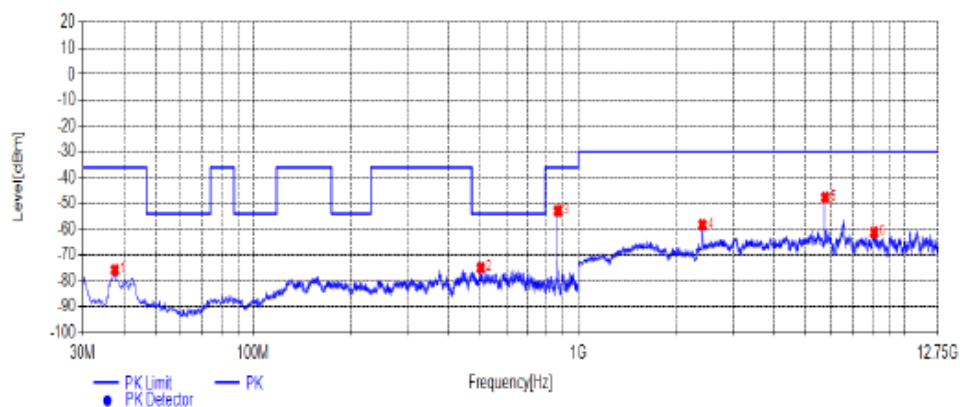
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.9556	-109.75	40.89	100	115	PK	Horizontal	PASS
2	129.5419	-103.04	42.34	100	13	PK	Horizontal	PASS
3	375.0010	-100.49	39.29	100	274	PK	Horizontal	PASS
4	864.9490	-101.62	12.15	100	332	PK	Horizontal	PASS
5	2410.2821	-99.96	23.99	100	261	PK	Horizontal	PASS
6	5764.4029	-97.00	13.75	100	267	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	39.74	100	270	PK	Vertical	PASS
2	501.1262	-101.98	20.96	100	170	PK	Vertical	PASS
3	864.1728	-101.71	16.90	100	230	PK	Vertical	PASS
4	2407.9316	-99.99	28.22	100	300	PK	Vertical	PASS
5	5745.5991	-97.11	17.65	100	10	PK	Vertical	PASS
6	8114.8730	-93.41	31.21	100	0	PK	Vertical	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

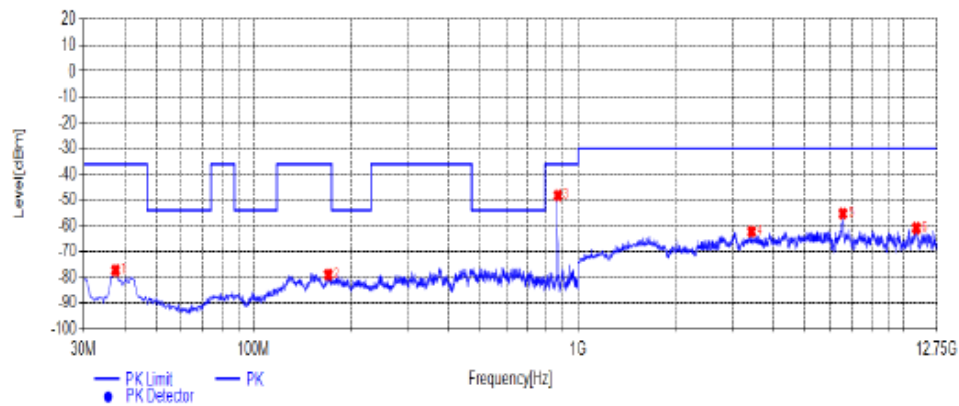
2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

865.0-868.0MHz:

Channel Bandwidth: 1MHz(865.5MHz)

Horizontal

Test Graph



Suspected List

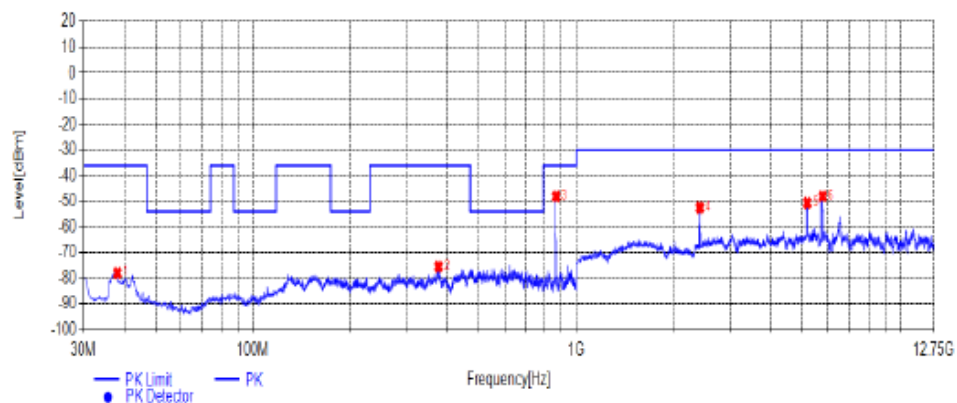
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	41.30	100	100	PK	Horizontal	PASS
2	169.9020	-103.35	43.05	100	290	PK	Horizontal	PASS
3	865.7251	-101.43	12.24	100	270	PK	Horizontal	PASS
4	3425.6851	-97.01	32.26	100	250	PK	Horizontal	PASS
5	6549.4599	-88.67	25.29	100	70	PK	Horizontal	PASS
6	11017.7035	-88.79	30.88	100	80	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	38.1496	-109.58	41.98	100	110	PK	Vertical	PASS
2	375.5831	-100.78	39.50	100	20	PK	Vertical	PASS
3	865.7251	-101.43	12.11	100	160	PK	Vertical	PASS
4	2412.6325	-99.93	22.52	100	40	PK	Vertical	PASS
5	5183.8368	-94.56	20.66	100	310	PK	Vertical	PASS
6	5783.2066	-97.15	18.03	100	40	PK	Vertical	PASS

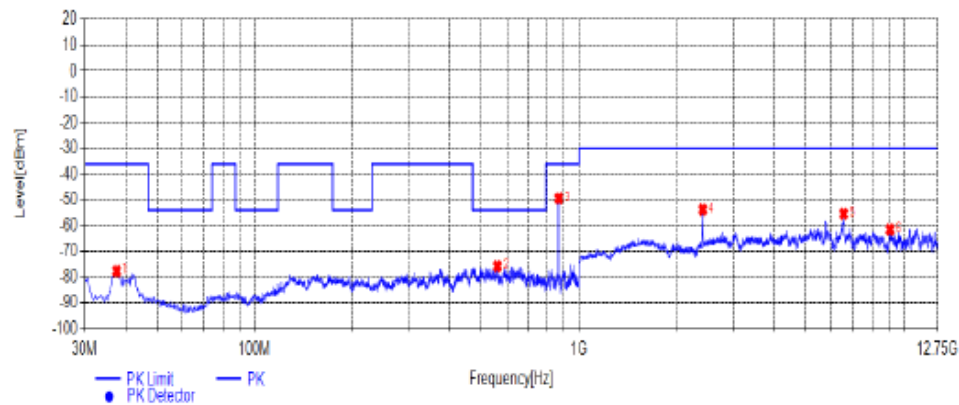
Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 1MHz(866.5MHz)

Horizontal

Test Graph



Suspected List

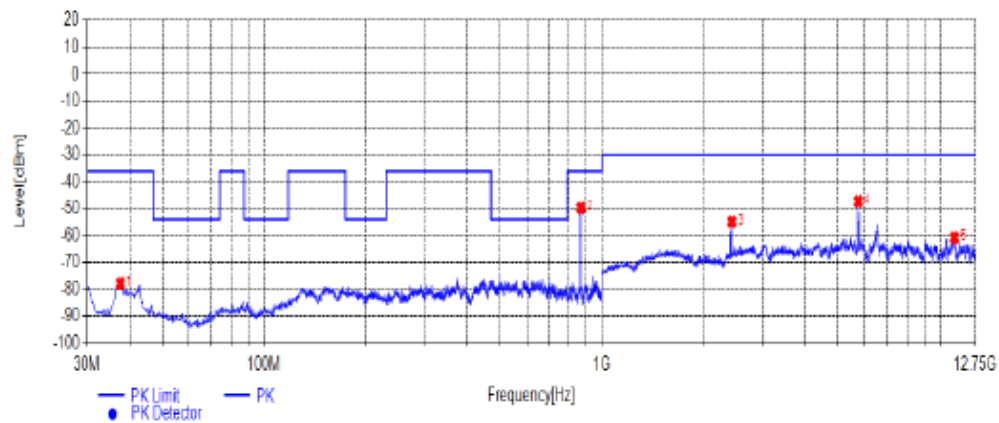
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	41.68	100	290	PK	Horizontal	PASS
2	560.6961	-101.41	21.74	100	250	PK	Horizontal	PASS
3	865.9192	-101.38	13.39	100	30	PK	Horizontal	PASS
4	2410.2821	-99.96	23.76	100	200	PK	Horizontal	PASS
5	6551.8104	-88.66	25.44	100	310	PK	Horizontal	PASS
6	9097.3695	-90.41	31.50	100	100	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	41.83	100	200	PK	Vertical	PASS
2	866.1132	-101.33	13.66	100	270	PK	Vertical	PASS
3	2426.7353	-99.76	24.85	100	230	PK	Vertical	PASS
4	5740.8982	-97.36	17.29	100	60	PK	Vertical	PASS
5	11048.2597	-88.55	30.81	100	250	PK	Vertical	PASS

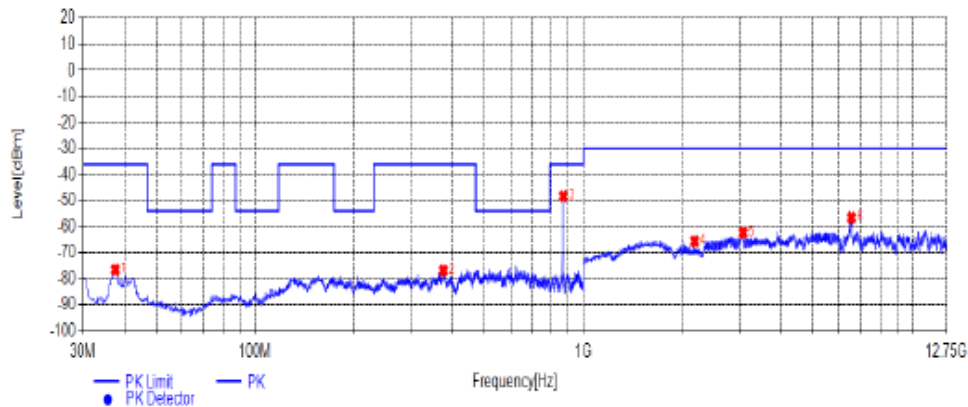
Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 1MHz(867.5MHz)

Horizontal

Test Graph



Suspected List

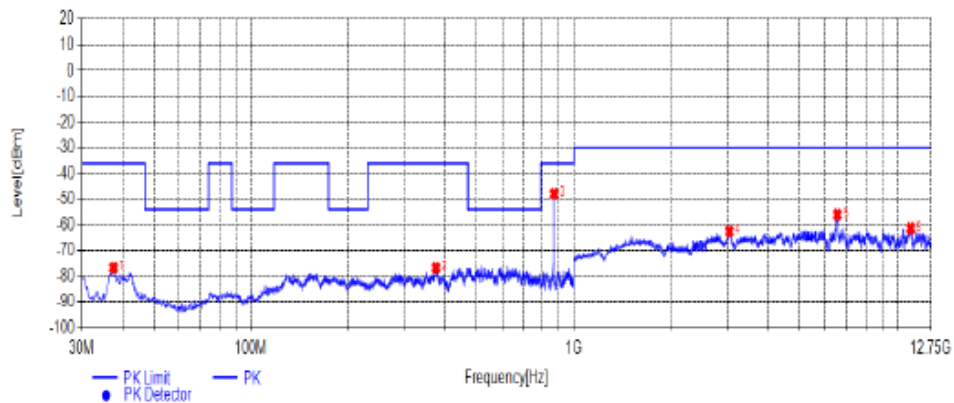
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	40.53	100	230	PK	Horizontal	PASS
2	375.1950	-100.58	40.85	100	100	PK	Horizontal	PASS
3	867.2775	-101.03	12.31	100	190	PK	Horizontal	PASS
4	2179.9360	-100.80	35.57	100	40	PK	Horizontal	PASS
5	3051.3623	-95.88	32.30	100	180	PK	Horizontal	PASS
6	6554.1608	-89.18	26.59	100	330	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	40.88	100	170	PK	Vertical	PASS
2	374.8070	-100.81	40.74	100	100	PK	Vertical	PASS
3	867.2775	-101.03	11.90	100	190	PK	Vertical	PASS
4	3030.8062	-96.26	32.55	100	0	PK	Vertical	PASS
5	6551.8104	-88.86	26.06	100	310	PK	Vertical	PASS
6	11036.5073	-88.64	31.31	100	80	PK	Vertical	PASS

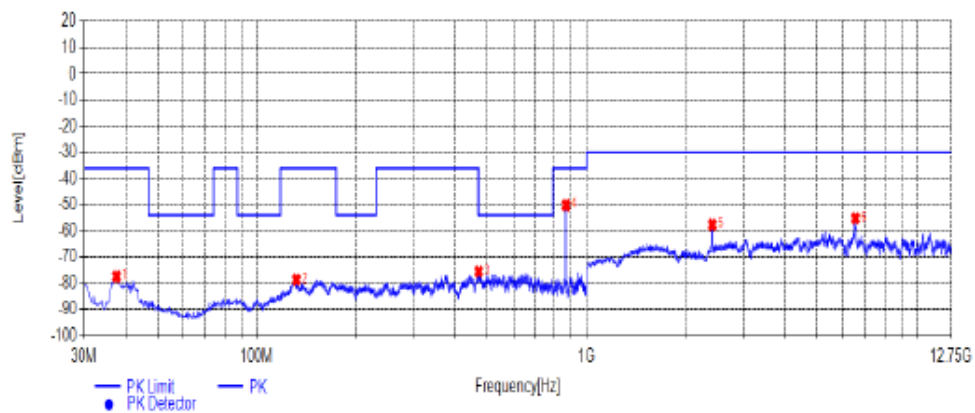
Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 2MHz(866.0MHz)

Horizontal

Test Graph



Suspected List

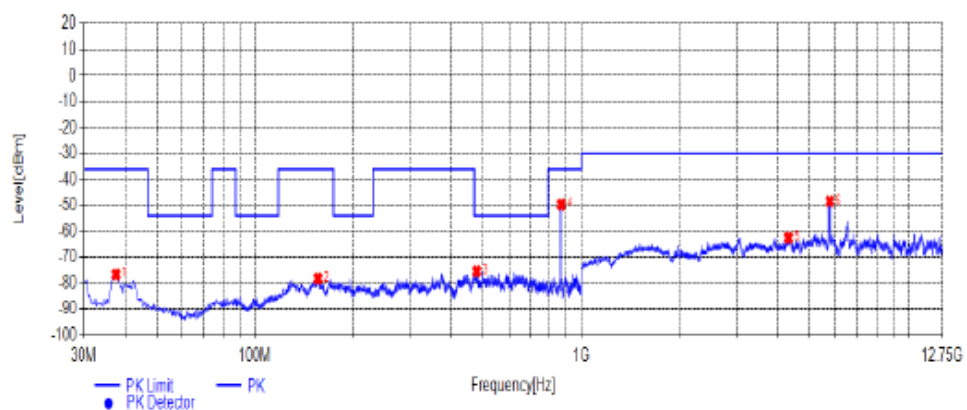
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	41.47	100	110	PK	Horizontal	PASS
2	132.0644	-103.64	42.71	100	270	PK	Horizontal	PASS
3	471.2442	-101.71	21.53	100	220	PK	Horizontal	PASS
4	865.5311	-101.48	14.17	100	210	PK	Horizontal	PASS
5	2407.9316	-99.99	27.65	100	40	PK	Horizontal	PASS
6	6540.0580	-89.72	25.23	100	240	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	40.64	100	170	PK	Vertical	PASS
2	156.1252	-102.47	42.25	100	190	PK	Vertical	PASS
3	478.2296	-101.54	21.44	100	130	PK	Vertical	PASS
4	866.6953	-101.18	13.65	100	110	PK	Vertical	PASS
5	4309.4619	-97.18	32.50	100	350	PK	Vertical	PASS
6	5771.4543	-97.06	18.42	100	270	PK	Vertical	PASS

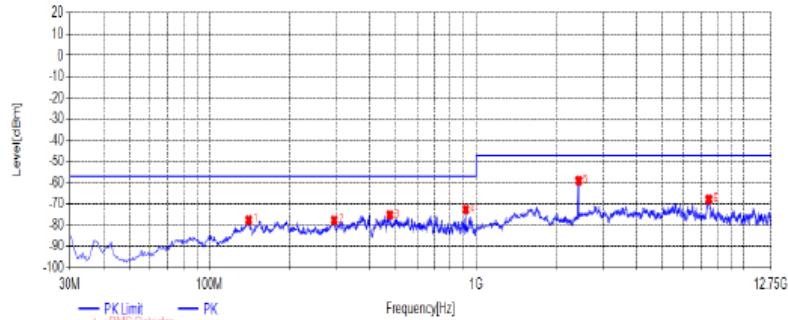
Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Test Mode: Standby (2MHz 866.0MHz)

Horizontal

Test Graph



Suspected List

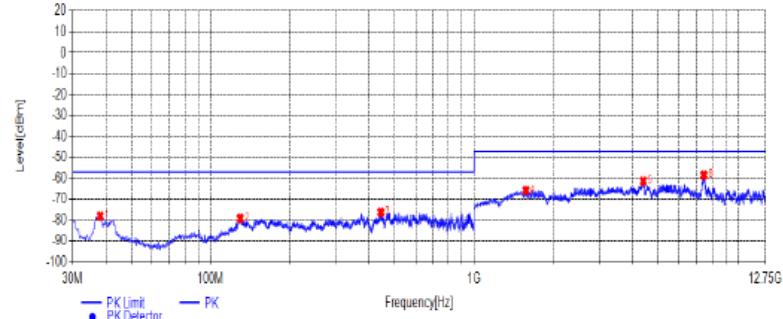
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	140.2361	-101.60	20.61	150	120	PK	Horizontal	PASS
2	293.9456	-101.70	20.69	150	250	PK	Horizontal	PASS
3	475.2141	-100.39	18.24	150	180	PK	Horizontal	PASS
4	913.4414	-103.21	15.60	150	340	PK	Horizontal	PASS
5	2424.6699	-99.79	12.03	150	80	PK	Horizontal	PASS
6	7455.6823	-90.79	20.79	150	170	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading (dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	38.1496	-109.58	21.06	150	230	PK	Vertical	PASS
2	129.5419	-103.04	22.22	150	340	PK	Vertical	PASS
3	443.3027	-101.75	19.21	150	70	PK	Vertical	PASS
4	1675.8652	-100.67	18.82	150	100	PK	Vertical	PASS
5	4391.7283	-95.00	14.34	150	70	PK	Vertical	PASS
6	7449.6899	-91.03	11.45	150	210	PK	Vertical	PASS

Note:1. Result (dBμV/m) = Reading (dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Remark:

- (1) Corrected Power (dBm) = SG O/P-Cable + Ant Gain
- (2) Measuring frequencies from 30 MHz to the 12.75GHz.
- (3) Data of measurement within this frequency range shown "--" in the table above means the reading of emissions are attenuated more than 20dB below the permissible limits or the field strength is too small to be measured.

Note: All the modes have been tested and recorded worst mode in the report.

16. BLOCKING (REVEIVER REQUIREMENTS)

16.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.9.2.

The blocking levels at the specified frequency offsets shall be equal to or greater than the limits in the following Table, except at frequencies where spurious responses are found.

Blocking level parameters for RX category 1

Requirement	Limits
	Receiver category 1
Blocking at ± 2 MHz from Centre Frequency	≥ -20 dBm
Blocking at ± 10 MHz from Centre Frequency	≥ -20 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -20 dBm

Blocking level parameters for RX category 1.5

Requirement	Limits
	Receiver category 1.5
Blocking at ± 2 MHz from Centre Frequency	≥ -43 dBm
Blocking at ± 10 MHz from Centre Frequency	≥ -33 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -33 dBm

Blocking level parameters for RX category 2

Requirement	Limits
	Receiver category 2
Blocking at ± 2 MHz from Centre Frequency	≥ -69 dBm
Blocking at ± 10 MHz from Centre Frequency	≥ -44 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -44 dBm

Blocking level parameters for RX category 3

Requirement	Limits
	Receiver category 3
Blocking at ± 2 MHz from Centre Frequency	≥ -80 dBm
Blocking at ± 10 MHz from Centre Frequency	≥ -60 dBm
Blocking at ± 5 % of Centre Frequency or 15 MHz, whichever is the greater	≥ -60 dBm

NOTE: For equipment using CCA whatever is the receiver category, steps 1 to 4 shall be repeated with signal generator A level adjusted +13 dB higher than in the measurements in clause 5.18.6.4.

16.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.18.6 for the measurement method.

16.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Mode: Receiving

This receiver is belongs to RX category 2

863.0-865.0MHz:

Channel Bandwidth: 1MHz

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
863.5MHz	863.5+2	-55	At least -69	Pass
863.5MHz	863.5+10	-36	At least -44	Pass
863.5MHz	863.5+15	-36	At least -44	Pass

Channel Bandwidth: 1MHz

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
864.5MHz	864.5+2	-59	At least -69	Pass
864.5MHz	864.5+10	-39	At least -44	Pass
864.5MHz	864.5+15	-38	At least -44	Pass

Channel Bandwidth: 2MHz

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
864.0MHz	864.0+2	-58	At least -69	Pass
864.0MHz	864.0+10	-39	At least -44	Pass
864.0MHz	864.0+15	-39	At least -44	Pass

865.0-868.0MHz:**Channel Bandwidth: 1MHz**

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
865.5MHz	865.5+2	-60	At least -69	Pass
865.5MHz	865.5+10	-41	At least -44	Pass
865.5MHz	865.5+15	-39	At least -44	Pass

Channel Bandwidth: 1MHz

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
866.5MHz	866.5+2	-59	At least -69	Pass
866.5MHz	866.5+10	-40	At least -44	Pass
866.5MHz	866.5+15	-38	At least -44	Pass

Channel Bandwidth: 1MHz

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
867.5MHz	867.5+2	-62	At least -69	Pass
867.5MHz	867.5+10	-40	At least -44	Pass
867.5MHz	867.5+15	-39	At least -44	Pass

Channel Bandwidth: 2MHz

Wanted Signal Channel (MHz)	Unwanted Signal Frequency (MHz)	Blocking level (dBm)	Limit (dBm)	Result
866.0MHz	866.0+2	-61	At least -69	Pass
866.0MHz	866.0+10	-38	At least -44	Pass
866.0MHz	866.0+15	-40	At least -44	Pass

17. RECEIVER SPURIOUS RADIATION

17.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.9.2.
The power of any spurious emission, radiated or conducted, shall not exceed the values given below. The limits are applicable to all receiver categories:

- -57dBm below 1 000 MHz;
- -47dBm above 1 000 MHz.

17.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.18.6 for the measurement method.

17.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Result:

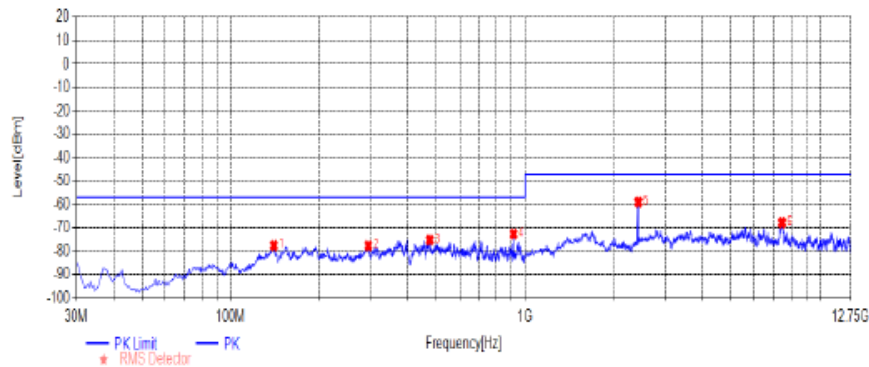
Test Mode: Receiving

863.0-865.0MHz:

Channel Bandwidth: 1MHz (863.5MHz)

Horizontal

Test Graph



Suspected List

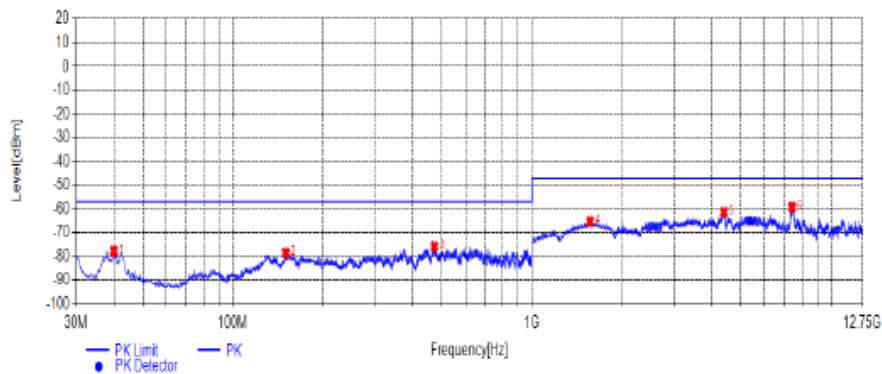
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	140.2361	-101.60	20.61	150	120	PK	Horizontal	PASS
2	293.9456	-101.70	20.69	150	250	PK	Horizontal	PASS
3	475.2141	-100.39	18.24	150	180	PK	Horizontal	PASS
4	913.4414	-103.21	15.60	150	340	PK	Horizontal	PASS
5	2424.6699	-99.79	12.03	150	80	PK	Horizontal	PASS
6	7455.6823	-90.79	20.79	150	170	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	40.0900	-108.01	20.75	150	230	PK	Vertical	PASS
2	150.4981	-102.94	21.59	150	80	PK	Vertical	PASS
3	472.9906	-101.74	18.81	150	330	PK	Vertical	PASS
4	1568.8138	-100.73	18.28	150	30	PK	Vertical	PASS
5	4391.7283	-95.00	14.39	150	0	PK	Vertical	PASS
6	7419.1338	-92.00	12.25	150	280	PK	Vertical	PASS

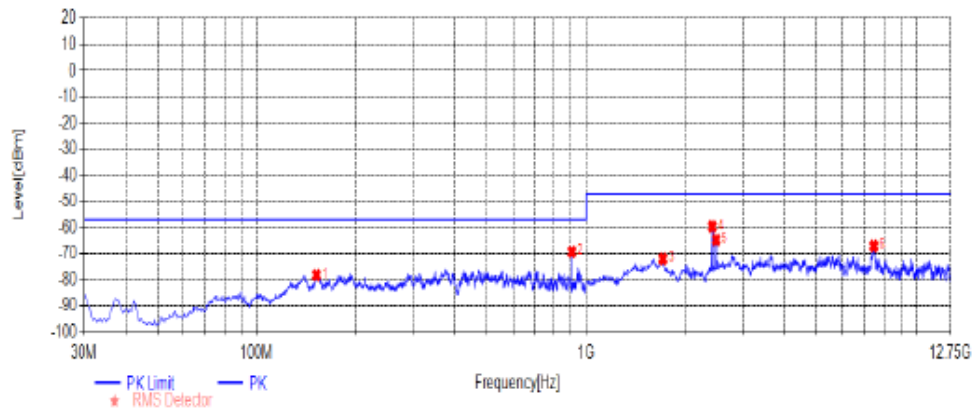
Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 2MHz (864.0MHz)

Horizontal

Test Graph



Suspected List

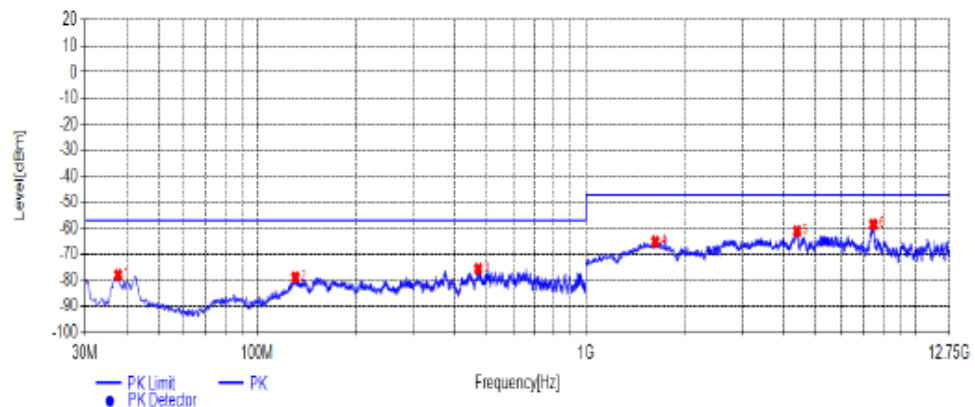
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	151.8808	-102.78	21.19	150	50	PK	Horizontal	PASS
2	904.5138	-105.00	12.27	150	40	PK	Horizontal	PASS
3	1709.9840	-97.37	24.91	150	330	PK	Horizontal	PASS
4	2415.2661	-99.90	12.61	150	190	PK	Horizontal	PASS
5	2481.0924	-99.31	17.70	150	40	PK	Horizontal	PASS
6	7465.0860	-90.96	20.06	150	170	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.7616	-109.89	20.94	150	180	PK	Vertical	PASS
2	130.7061	-103.04	21.86	150	210	PK	Vertical	PASS
3	470.4681	-101.71	18.54	150	50	PK	Vertical	PASS
4	1627.5755	-100.26	18.07	150	0	PK	Vertical	PASS
5	4391.7283	-95.00	14.09	150	70	PK	Vertical	PASS
6	7489.6479	-91.52	11.51	150	90	PK	Vertical	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

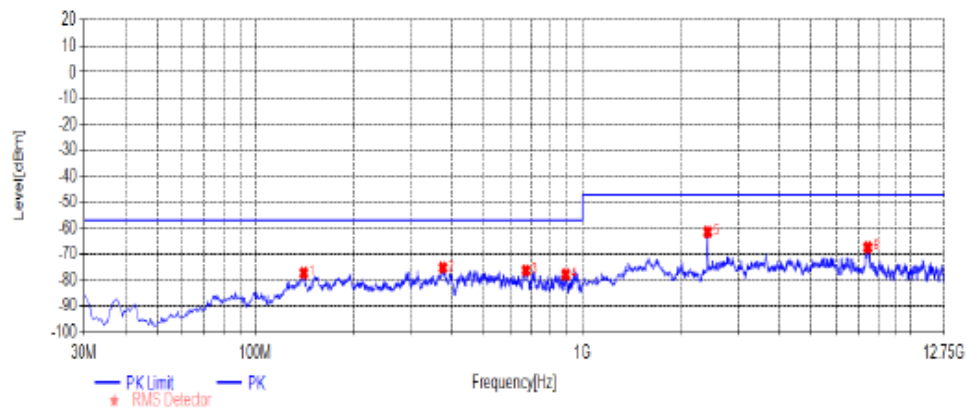
2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

865.0-868.0MHz:

Channel Bandwidth: 1MHz (865.5MHz)

Horizontal

Test Graph



Suspected List

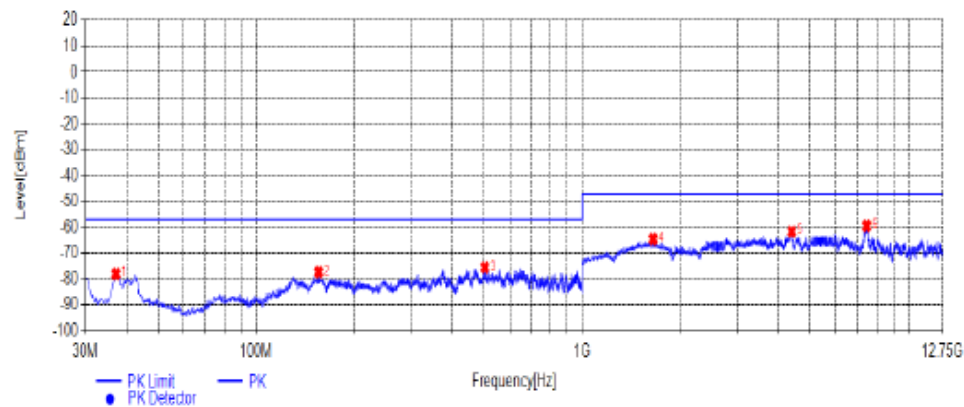
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	140.6243	-102.01	20.27	150	270	PK	Horizontal	PASS
2	375.0700	-100.02	18.21	150	230	PK	Horizontal	PASS
3	672.0088	-100.14	19.25	150	20	PK	Horizontal	PASS
4	888.2113	-100.50	20.83	150	340	PK	Horizontal	PASS
5	2415.2661	-99.90	14.42	150	340	PK	Horizontal	PASS
6	7446.2785	-90.81	20.40	150	320	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.1794	-110.33	21.04	150	120	PK	Vertical	PASS
2	155.5431	-102.37	20.35	150	150	PK	Vertical	PASS
3	502.6785	-102.21	18.52	150	260	PK	Vertical	PASS
4	1651.0802	-100.10	17.53	150	20	PK	Vertical	PASS
5	4398.7798	-94.63	14.66	150	290	PK	Vertical	PASS
6	7469.0918	-91.13	12.47	150	340	PK	Vertical	PASS

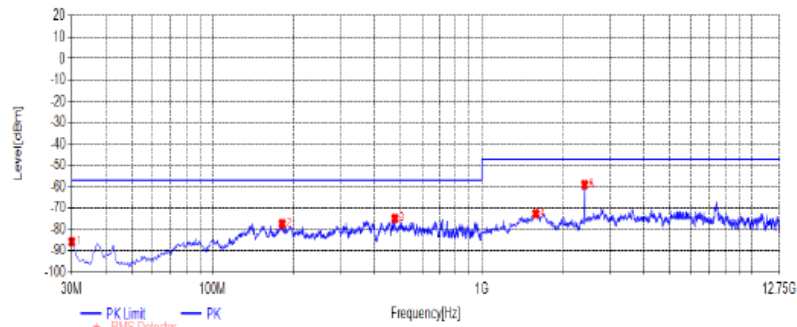
Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Channel Bandwidth: 2MHz (866.0MHz)

Horizontal

Test Graph



Suspected List

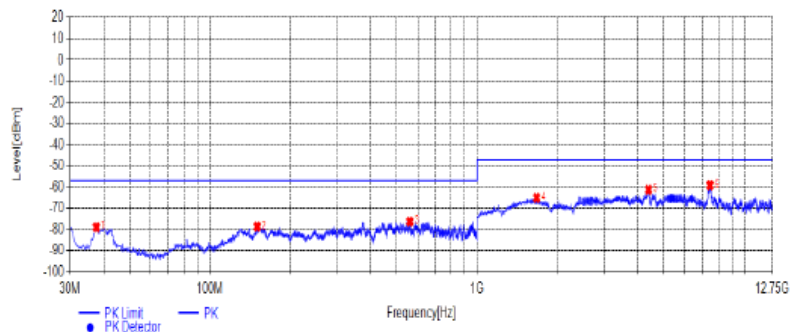
NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	30.0000	-116.14	28.84	150	130	PK	Horizontal	PASS
2	181.3806	-102.19	20.41	150	170	PK	Horizontal	PASS
3	475.9904	-100.84	17.87	150	90	PK	Horizontal	PASS
4	1587.7351	-98.38	25.61	150	290	PK	Horizontal	PASS
5	2410.5642	-99.96	12.08	150	240	PK	Horizontal	PASS
6	2410.5642	-99.96	12.08	150	240	PK	Horizontal	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Vertical

Test Graph



Suspected List

NO.	Frequency [MHz]	Factor [dB]	Margin [dB]	Height [cm]	Angle [°]	Detector	Polarity	Remark
1	37.5675	-110.04	22.02	150	100	PK	Vertical	PASS
2	150.4981	-102.94	21.72	150	240	PK	Vertical	PASS
3	560.6961	-101.41	19.30	150	160	PK	Vertical	PASS
4	1672.2344	-99.96	18.05	150	250	PK	Vertical	PASS
5	4391.7283	-95.00	14.23	150	160	PK	Vertical	PASS
6	7452.0404	-91.04	12.16	150	260	PK	Vertical	PASS

Note:1. Result (dBμV/m) = Reading(dBμV/m) + Factor (dB) .

2. Factor (dB) = Antenna Factor (dB/m) + Cable loss (dB) - Pre Amplifier gain (dB).

Remark:

- (1) Corrected Power (dBm) = SG O/P-Cable + Ant Gain
- (2) Measuring frequencies from 30 MHz to the 12.75GHz.
- (3) Data of measurement within this frequency range shown "--" in the table above means the reading of emissions are attenuated more than 20dB below the permissible limits or the field strength is too small to be measured.

Note: All the modes have been tested and recorded worst mode in the report.

18. RX sensitivity

18.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.14.2.
The sensitivity for receivers shall be below or equal to Table 32 level.

Table 32: Limits for Receiver sensitivity

$S = 10 \log RB_{kHz} - 4 \text{ dB}\mu V \text{ emf}; \text{ or}$ $S_p = 10 \log RB_{kHz} - 117 \text{ dBm}$ where: • S_p is the sensitivity in dBm. • RB is the declared receiver bandwidth in kHz.

Note:The receiver bandwidth RB shall be declared by the manufacturer. RB is the usually 3 dB receiver bandwidth selectivity.

18.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.14.3.3 for the measurement method.

18.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Result:

Test Mode: Receiving

863.0-865.0MHz:**Channel Bandwidth: 1MHz(863.5MHz)**

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
863.5	Modulation Signal	-87.32	-88.97	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 1MHz Operating Channel equipment with a 929KHz bandwidth.

Channel Bandwidth: 1MHz(864.5MHz)

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
864.5	Modulation Signal	-87.32	-89.69	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 1MHz Operating Channel equipment with a 929KHz bandwidth.

Channel Bandwidth: 2MHz(864.0MHz)

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
864.0	Modulation Signal	-84.26	-89.64	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 2MHz Operating Channel equipment with a 1881KHz bandwidth.

865.0-868.0MHz:**Channel Bandwidth: 1MHz(865.5MHz)**

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
865.5	Modulation Signal	-87.34	-90.15	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 1MHz Operating Channel equipment with a 924KHz bandwidth.

Channel Bandwidth: 1MHz(866.5MHz)

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
866.5	Modulation Signal	-87.37	-90.69	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 1MHz Operating Channel equipment with a 919KHz bandwidth.

Channel Bandwidth: 1MHz(867.5MHz)

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
867.5	Modulation Signal	-87.30	-89.63	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 1MHz Operating Channel equipment with a 934KHz bandwidth.

Channel Bandwidth: 2MHz(866.0MHz)

Frequency (MHz)	Test Signal	Sensitivity Limit (dBm)	Measured sensitivity (dBm)	Result
866.0	Modulation Signal	-84.53	-88.76	Pass

Note: 1) The receiver sensitivity should be measured with any Forward Error Correction (FEC) or Automatic Repeat reQuest (ARQ) function disabled.
 2) after demodulation, a raw data signal with a Bit Error Ratio of 10⁻³ without correction;
 3) the sensitivity for a 2MHz Operating Channel equipment with a 1768KHz bandwidth.

19. Clear Channel Assessment threshold

19.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.21.2.
The CCA threshold shall not exceed the limits given in Table 45.

Table 45: CCA radiated threshold limits

Parameter	Value
CCA threshold for EUT with e.r.p < 100 mW	15 dB above Rx sensitivity level limit as given in Table 32
CCA threshold for EUT with e.r.p from 100 mW to 500 mW	11 dB above Rx sensitivity level limit as given in Table 32
NOTE: The limits are based on an antenna gain of 0 dB relative to a dipole (i.e. +2,15 dBi) maximum. For other antenna gains different to 0 dB the limits shall be adjusted accordingly.	
EXAMPLE: With antenna gain of -5 dB, the CCA threshold shall be corrected and increased by 5 dB at the RF front end.	

19.2. Test Procedure

Please refer to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.21.2.3.3 for the measurement method.

19.3. Test Data

Environmental Conditions

Temperature:	24.2 °C
Relative Humidity:	54.6%
ATM Pressure:	100.0 kPa

Test Result:

Test Mode: Receiving

863.0-865.0MHz:**Channel Bandwidth: 1MHz(863.5MHz)****First measured**

Frequency (MHz)	Test Siganl	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
863.5	Modulation Signal	-73.12	No	-72.32	Pass

Second measured

Frequency (MHz)	Test Siganl	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
863.5	Modulation Signal	-73.21	No	-72.32	Pass

Channel Bandwidth: 1MHz(864.5MHz)**First measured**

Frequency (MHz)	Test Siganl	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
864.5	Modulation Signal	-73.25	No	-72.32	Pass

Second measured

Frequency (MHz)	Test Siganl	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
864.5	Modulation Signal	-73.17	No	-72.32	Pass

Channel Bandwidth: 2MHz(864.0MHz)**First measured**

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
864.0	Modulation Signal	-71.69	No	-69.26	Pass

Second measured

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
864.0	Modulation Signal	-71.84	No	-69.26	Pass

865.0-868.0MHz:**Channel Bandwidth: 1MHz(865.5MHz)****First measured**

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
865.5	Modulation Signal	-73.56	No	-72.34	Pass

Second measured

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
865.5	Modulation Signal	-73.45	No	-72.34	Pass

Channel Bandwidth: 1MHz(866.5MHz)**First measured**

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
866.5	Modulation Signal	-73.32	No	-72.37	Pass

Second measured

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
866.5	Modulation Signal	-73.45	No	-72.37	Pass

Channel Bandwidth: 1MHz(867.5MHz)**First measured**

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
867.5	Modulation Signal	-73.26	No	-72.30	Pass

Second measured

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
867.5	Modulation Signal	-73.38	No	-72.30	Pass

Channel Bandwidth: 2MHz(866.0MHz)**First measured**

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
866.0	Modulation Signal	-71.69	No	-69.53	Pass

Second measured

Frequency (MHz)	Test Signal	CCA threshold	Presence of unexpected EUT signal	Limit(dB m)	Result
866.0	Modulation Signal	-71.75	No	-69.53	Pass

NOTE:

- 1) The presence of unexpected EUT transmission is a test failure;
- 2) CCA threshold just signal generator level when EUT begins transmission .

20. Clear Channel Assessment threshold

20.1. Standard Applicable

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.21.3.
Parameters are shown in Table 48 and their limits shall not be exceeded.

Table 48: Reference limits for Polite Spectrum Access timing parameters

Parameter	Limit	Notes
Minimum CCA interval	160 μ s	Minimum CCA listening period
Minimum deferral period	CCA interval	Minimum value of the deferral interval
Minimum unit of deferral period	See note 1	Smallest interval between two adjacent deferral periods
Dead Time	CCA declared time by the vendor but not exceeding 5 ms	Maximum time between the end of a listening interval and the start of a transmission
Maximum Transmission Duration $T_{on\ Max}$	1 s	For a single Transmission
Maximum Transmission Duration $T_{on\ Max}$	4 s	For a Transmission dialogue or a polling sequence
Max T_{cum_on} over 1 hour	100 s / 1 h per 200 kHz spectrum	Maximum allowed Cumulative On Time over a 200 kHz portion of spectrum per hour
Minimum T_{off_min} on the same Operating Frequency	100 ms	The minimum T-off time period where a specific transmitter shall remain off after a transmission on the same Operating Frequency
NOTE 1: The minimum unit of deferral period is declared by the manufacturer.		
NOTE 2: T_{DIS} is declared by the manufacturer without exceeding 5 ms.		
NOTE 3: Longer accumulated transmission time is possible by implementing more AFA channels.		

20.3. Test Data

The manufacturer declare as follows

Parameter	Test Value	Limit	Result
Minimum CCA interval	264 μ s	160 μ s	PASS
Maximum dead time	130 μ s	CCA declared time by the vendor but not exceeding 5 ms	PASS
Method employed to randomize timing of re-transmission attempts	Random binary backoff	/	PASS
Units of the deferral period	52 μ s	See note 1	PASS
Minimum and maximum values of the deferral random period	Mini 264 μ s, Max53356 μ s	Mini 160 μ s	PASS
Maximum transmission duration time $T_{on\ max}$	21ms	1 s	PASS
Minimum T_{off_min} on the same Operating Frequency	100 ms	100 ms	PASS

18. LIST MEASURING EQUIPMENT

RF output power&PSD&OOB&OBW &Hopping &Duty Cycle, Tx-sequence, Tx-gap&Adaptively& ReceiverBlocking& Centre frequencies & TPC						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibration Date	Calibration Due Date
1	Spectrum Analyzer	Agilent	N9020A	MY48010425	2021/09/19	2022/09/18
2	Vector Signal generator	Agilent	N5181A	MY49060502	2021/07/13	2022/07/12
3	Signal generator	Agilent	E4421B	3610AO1069	2021/09/19	2022/09/18
4	4 Ch. Simultaneous Sampling 14 Bits 2 MS/s	Agilent	U2531A	TW53323507	2021/09/19	2022/09/18
5	X-series USB Peak and Average Power Sensor	Agilent	U2021XA	MY5365004	2021/09/19	2022/09/18
6	Climate Chamber	ESPEC	EL-10KA	A20120523	2021/09/19	2022/09/18
7	Spectrum Analyzer	R&S	FSV40	100019	2021/07/23	2022/07/22
8	Universal Radio Communication	Rohde&Schwarz	CMU200	114353	2021/09/19	2022/09/18
9	Wireless Communication Tester	Rohde&Schwarz	CMW500	125408	2021/07/13	2022/07/12
10	Test Control Unit	Tonscend	JS0806-1	178060067	2021/07/21	2022/07/20
11	Automated filter bank	Tonscend	JS0806-F	19F8060177	2021/07/23	2022/07/22
12	EMI Test software	Tonscend	JS1120-1	Ver 2.6.8.0518	/	/
13	EMI Test software	Tonscend	JS1120-3	Ver 2.5.77.0418	/	/

Transmitter spurious emissions & Receiver spurious emissions						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibration Date	Calibration Due Date
1	EMI Test Receiver	ROHDE & SCHWARZ	ESCI 7	101102	2021/09/19	2022/09/18
2	Spectrum Analyzer	Agilent	N9020A	MY480 10425	2021/09/19	2022/09/18
3	Spectrum Analyzer	R&S	FSV40	100019	2021/07/23	2022/07/22
4	By-log Antenna	SCHWARZBECK	VULB9163	000976	2021/07/23	2022/07/22
5	Double Ridged Horn Antenna (1~18GHz)	SCHWARZBECK	BBHA 9120D	01622	2021/09/19	2022/09/18
6	Horn Antenna (18GHz~40GHz)	Schwarzbeck	BBHA9170	791	2021/09/19	2022/09/18
7	Amplifier (30MHz~1GHz)	Schwarzbeck	BBV 9743	#202	2021/07/23	2022/07/22
8	Amplifier (1GHz~18GHz)	Taiwan Chengyi	EMC051845 B	980355	2021/07/23	2022/07/22
9	Amplifier (26.5GHz~40GHz)	Schwarzbeck	BBV9179	9719-0 25	2021/07/23	2022/07/22
10	High-Pass Filter	K&L	9SH10-2700 /X12750-O/ O	KL1420 31	2021/07/23	2022/07/22
11	High-Pass Filter	K&L	41H10-1375/ U12750-O/O	KL1420 32	2021/07/23	2022/07/22
12	High pass filter	Compliance Direction systems	BSU-6	34202	2021/07/23	2022/07/22
13	RF Cable	HUBER+SUHNER	RG214	N/A	2021/07/23	2022/07/22
14	EMI Test software	Tonscend	JS32-RE	Ver 2.5.1.8	/	/

19. TEST SETUP PHOTOGRAPHS

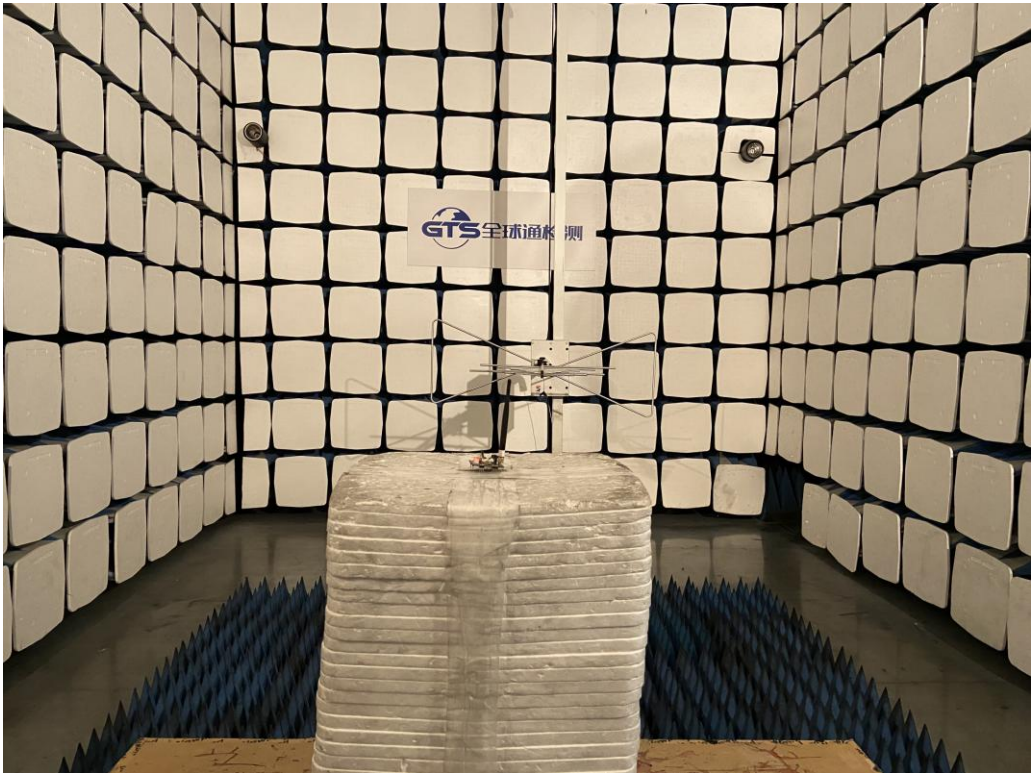


Fig. 1

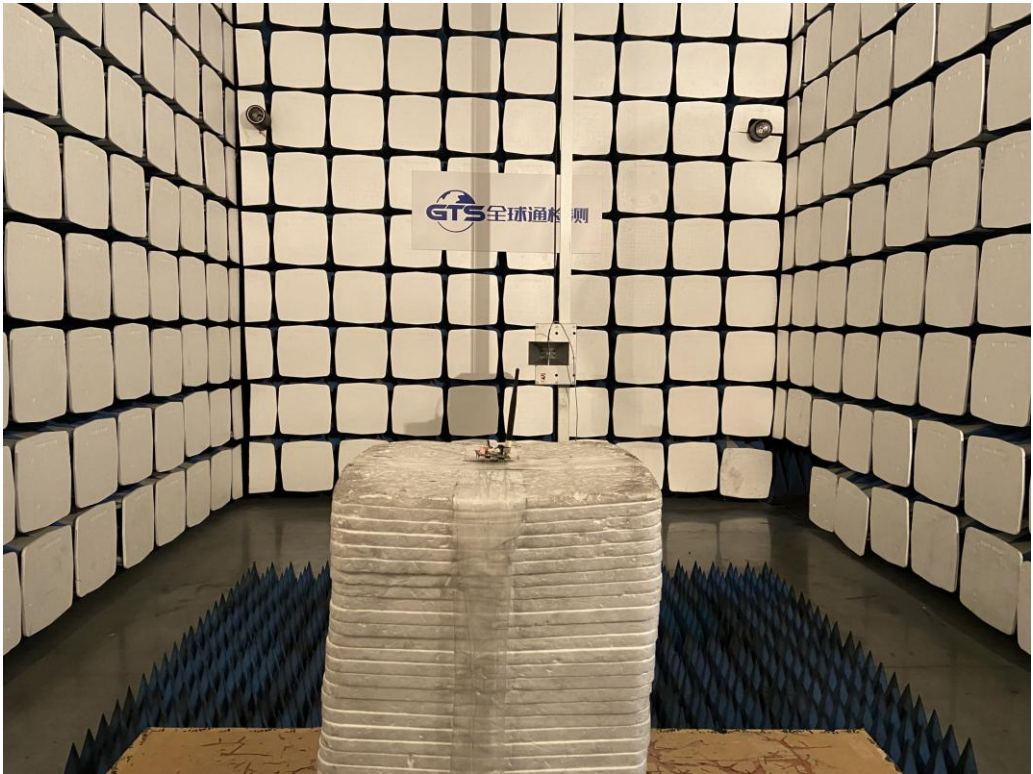


Fig. 2

20. EXTERNAL AND INTERNAL PHOTOS OF THE EUT

Reference to the test report No. GTS20211129024-1-3.

----- THE END OF REPORT-----